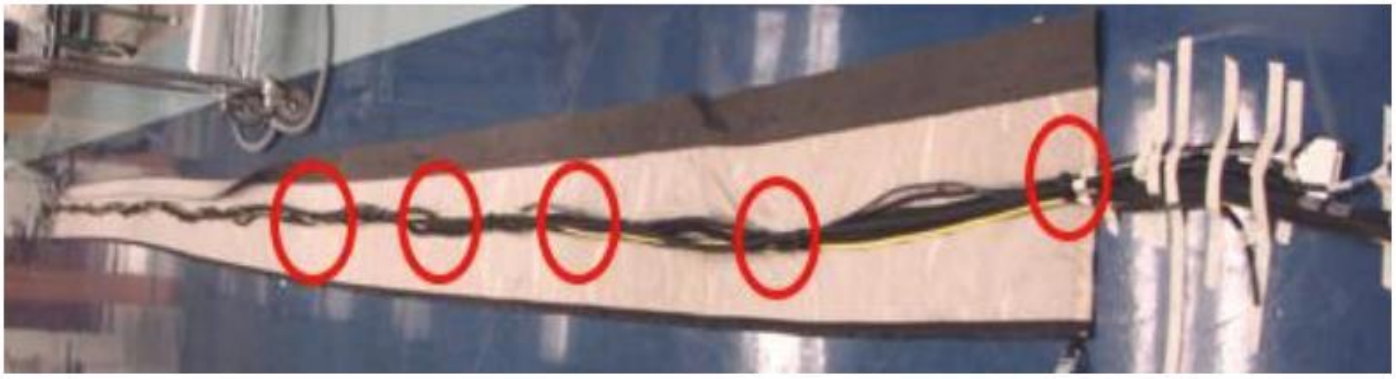
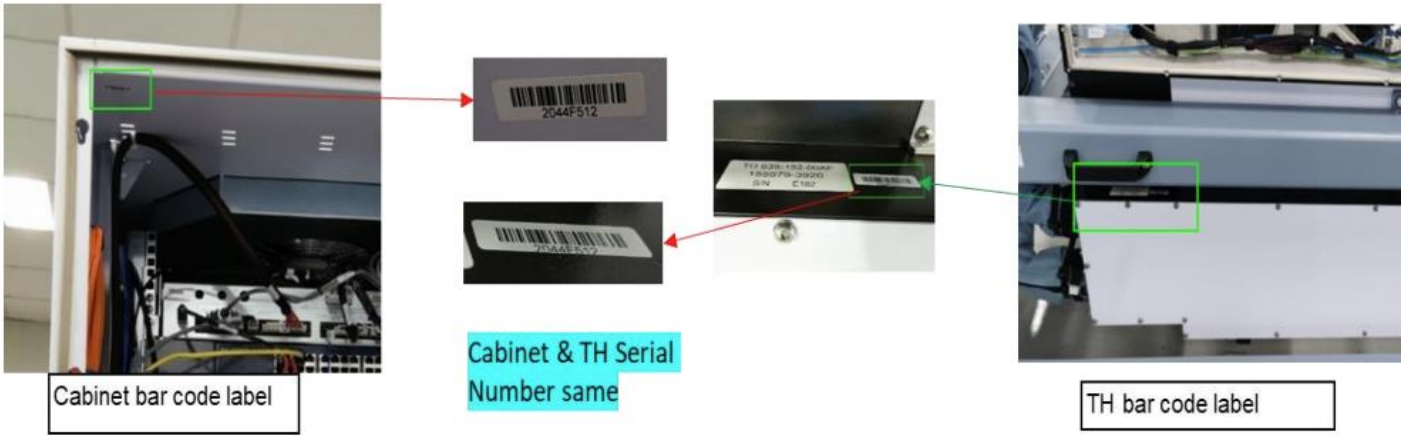
Build Overview:





CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	651-219-12	CABLE SOCK UF2 12-SLOT EMI CBL CVR BLACK	2
2	651-816-04	CABLE UF2 12SLOT SB25 SMCPWR INPUT TELECOM WITH CONN	1
3	655-540-03	CABLE UF2_SC_TH_EMO_LED_INTLK_A SM CONN TELECOM WITH	1
4	657-472-03	CABLE UF2 SC TH SMCLON_PRESS_DISCON_THA TELECOM WITH	1
5	651-111-02	CABLE UF2 12SLT LCL ENET TH TO SC TELECOM WITH CONN	1
6	655-233-01	CABLE UF2 12 SLOT LCL EXTERNAL CONTROL TELECOM WITH CONN	1
7	660-546-00	PLASTIC TUBE 6MM ODX1MMX 12MT BLACK HOSE PIPE OR TUBE	1
8	813-929-00	HDW VELCRO STRP 5/8	50
9	655-181-03	CABLE UF2 SUPP BRD FO LONG W TAB OPTICAL FIBER	2
10	655-185-03	CABLE UF2 INST FO TH TO SC W TAB OPTICAL FIBER	12
11	663-760-00	PLASTIC, DRAGON 12 SLOT FIBER OPTIC LABEL INFORMATION (TH	1
12	634-324-04	PLASTIC, UF2 SPLITLOOM TH TO SC .75, HOSE, PIPE OR TUBE	1
13	710-048-10	TRANCEIVERS PIN SFP+ MULTI 10GBS HOT PLUG DUPLEX CN	1
14	650-824-00	CABLE TH SUPPLY 48V WITHOUT CONN	1
15	650-825-00	CABLE TH RETURN 48V WITHOUT CONN	1
16	651-109-04	CABLE T/H 12SLT GROUND WIRE TELECOM WITH CONN	1
17	651-112-03	CABLE, UF2 12SLOT GND SC TO TH TERMBLK, TELECOM, WITH CONN	1
18	603-096-01	NUT HEX SS M8X1.25 WSHR W/TTH 13MM WDH	1
19	813-929-01	HDW VELCRO STRP 5/8	7
20	603-096-00	NUT HEX SS M5X0.8 W/WSHR W/TTH 8MM WDH	3
21	603-093-45	STANDOFF 10MM HEX M-F M5X.8X20MML SS	1
22	470-502-05	WASHER LK SPLT M5, 9.2MM OD, S	1
23	660-866-50	MECHANICAL, UF2 SC BUSBAR COVER, LEFT, FOR SEMI TEST	1
24	660-866-51	MECHANICAL, UF2 SC BUSBAR COVER, RIGHT, FOR SEMI TEST	1
25	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8x12MM L W/CONE	11
26	660-866-52	MECHANICAL, UF2 SC BUSBAR COVER, TOP, FOR SEMI TEST	1
27	479-133-18	LBL DANGER HAZ VLT 2.75X1.35"	2
28	938-102-00	TIE WRAP 6.5 X 3/16"	2
29	630-258-40	ELEC-MECH EMI CBL CVR SIGNAL BLACK FOR SEMI TEST	1
30	661-717-00	PLASTIC, UFPLUS SYSTEM MODEL, LABEL	1
31	479-133-42	CHINA ROHS LABEL	1
32	FLX-CON-TAP H00269	MASKING TAPE	1
33	FLX-PKG-LBL-H00081	Polyester, 40mm, 10mm, white,N;Label	1

METHOD:

Flow chart of operation steps

SIC preparation
↓
Preparation Tools
↓
Prepare Materials
↓
Complete Operation
↓
Inventory Material
↓
Operation Check
↓
Inventory tools

MACHINE:

Tool name	Torque value (pcs)
5/16 open torque wrench	0.8 N·m (7 in-lbs)
13mm socket wrench	13.5 N·m
PH2 Phillips torque screwdriver	3.3 N·m
10mm socket wrench	3.3 N·m
PH2 Phillips torque screwdriver	1.6 N·m
8mm socket wrench	1.6 N·m
10mm socket wrench	5.6 N·m
flat-blade screwdriver	0.19 N·m
13mm socket wrench	3.3 N·m
PH2 Phillips torque screwdriver	0.7 N·m
Duct Aid	NA
Measuring Tape	NA
Masking tape	NA
Tweezers	NA
Cutter	NA
Scissor	NA

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

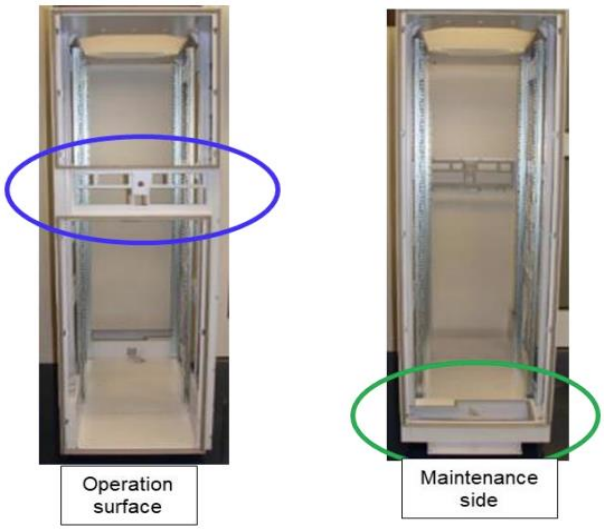
MACHINE: N/A

METHOD:

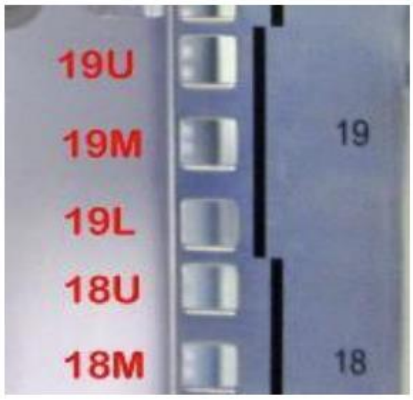
Operations step

1.1 The operating direction of the cabinet as shown in the figure-the operating direction of the cabinet panel is also defined as the front of the cabinet.

1.2 The cabinet maintenance surface as shown in the figure-the direction of the cabinet facilities on the bottom panel is also defined as the back of the cabinet.



1.3 The position of the guide rail is defined by numbers, which are printed on the guide rail. The positions of the numbers are U (top), M (middle), and L (bottom).



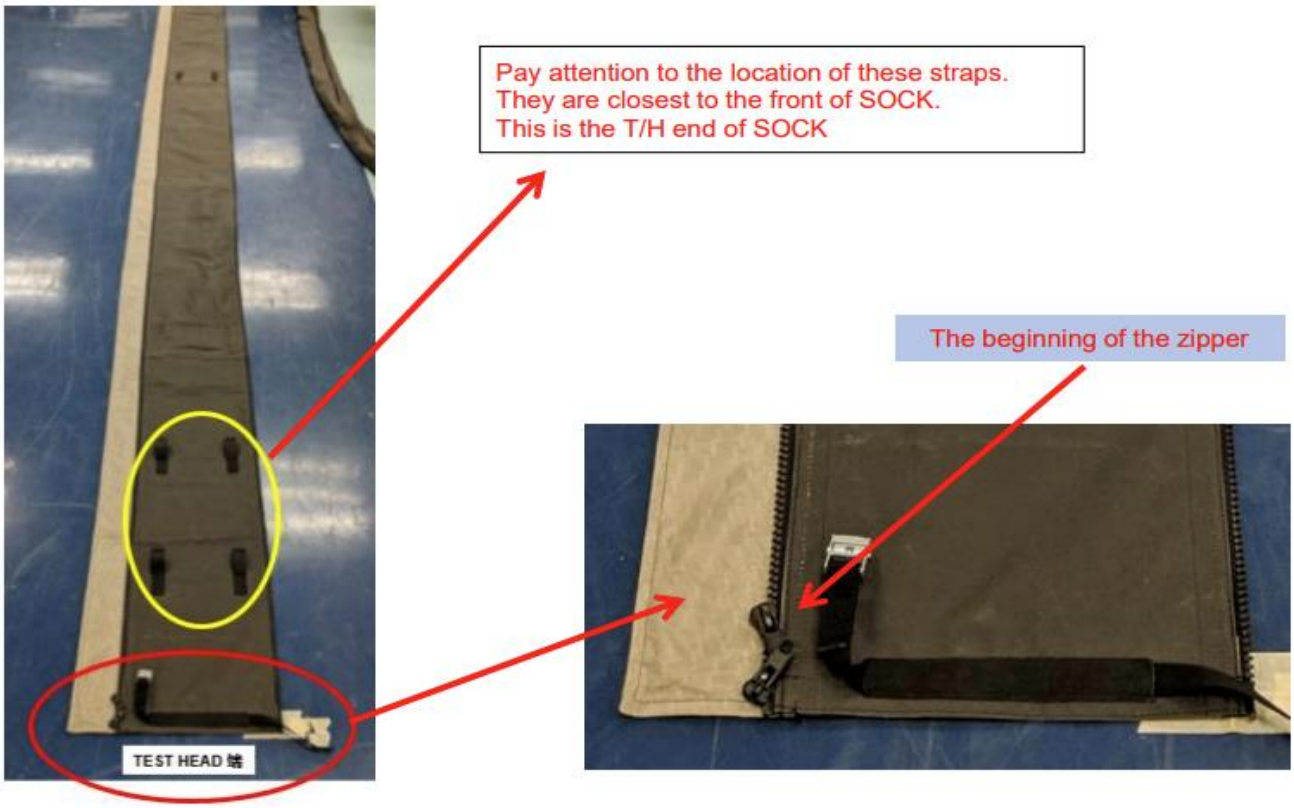
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

1.4 As shown in the picture, it is the T/H end of the zipper. The direction of the thread sleeve here will be connected to the T/H.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	651-219-12	CABLE SOCK UF2 12-SLOT EMI CBL CVR BLACK	1

MACHINE:

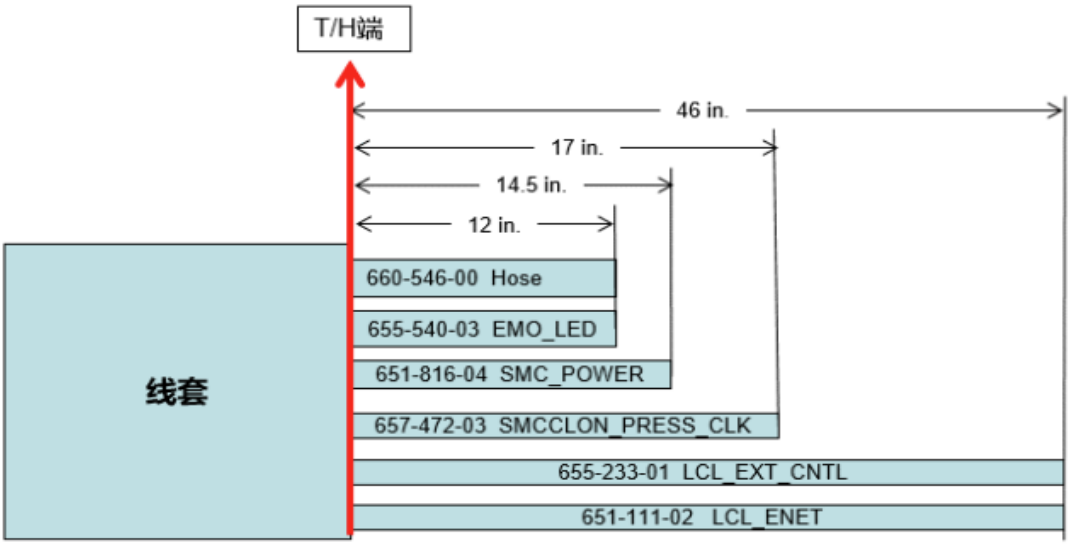
FLX-CON-TAP-H00269 MASKING TAPE

METHOD:

2.1 Place one EMI cable sleeve (651-219-12) on a flat surface with the shielding surface facing upward.

2.2 As shown in the figure, stick a piece of masking tape on the side of the T/H end of the wire sleeve for marking in the next steps.

2.3 Mark the masking tape according to the layout below.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

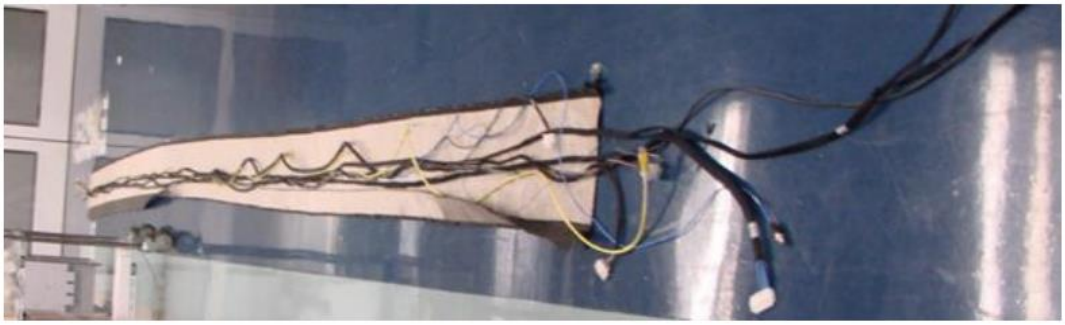
ITEM	P/N	DESCRIPTION	QTY
2	651-816-04	CABLE UF2 12SLOT SB25 SMCPWR INPUT TELECOM WITH	1
3	655-540-03	CABLE UF2_SC_TH_EMO_LED_INTLK_A SM CONN TELECOI	1
4	657-472-03	CABLE UF2 SC TH SMCLON_PRESS_DISCON_THA TELECOI	1
5	651-111-02	CABLE UF2 12SLT LCL ENET TH TO SC TELECOM WITH CON	1
6	655-233-01	CABLE UF2 12 SLOT LCL EXTERNAL CONTROL TELECOM W	1
7	660-546-00	PLASTIC TUBE 6MM ODX1MMX12MT BLACK HOSE PIPE OR	1

MACHINE:

METHOD:

3.1 Mark the two ends of a UF2 12 Slot SB25 SMCPWR Input cable (651-816-04) as shown below.
A. TH end: "[X] THA DISCPNL J3"
B. Cabinet side: "[X] SMC J9"

3.2 According to the size on the tape, lay the following cables on the wire. The length is to the edge of the thread sleeve.
A.651-816-04 UF2 12 Slot SB25 SMCPWR Input (1 piece)
B.655-540-03 UF2 SC TH EMO LED INTLK (1 piece)
C.657-472-03 UF2 SC TH SMCLON PRESS_DISCON (1 piece)
D.651-111-02 UF2 12 Slot LCL ENT TH to SC (1 piece)
E.655-233-01 UF2 12 Slot LCL External Control (1 piece)
F.660-546-00 Air Hose (1 piece)



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

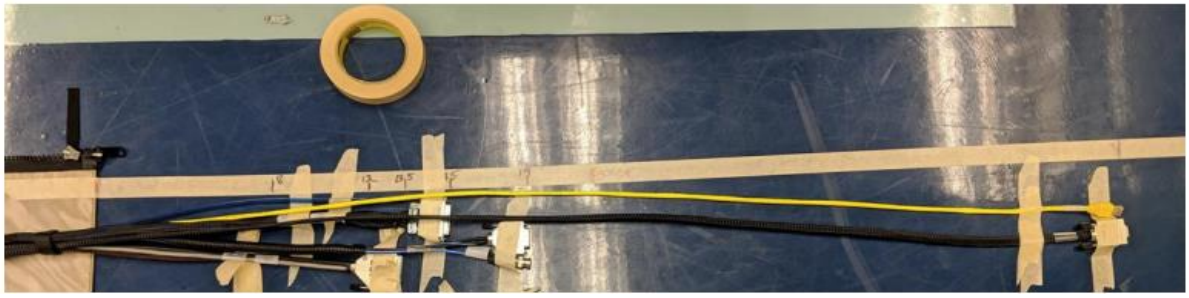
ITEM	P/N	DESCRIPTION	QTY
8	813-929-00	HDW VELCRO STRP 5/8	14

MACHINE:

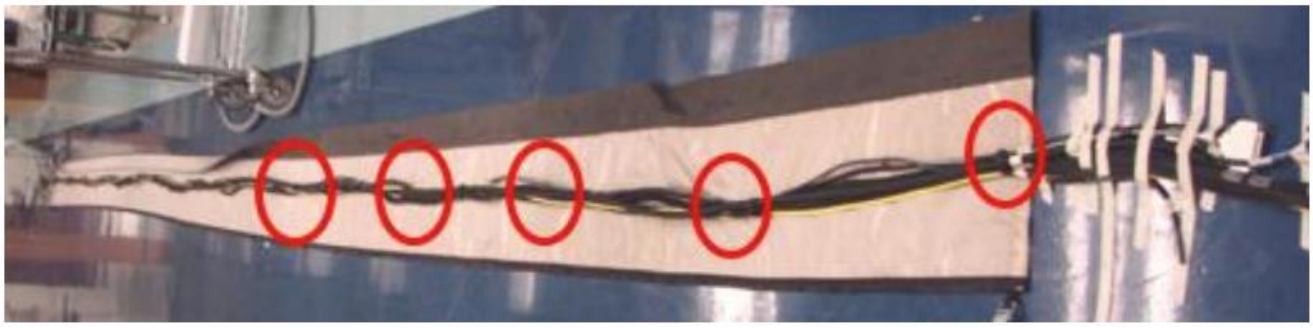
N/A

METHOD:

3.3 Fix the wires with masking tape.



3.4 Use cable ties 813-929-00 (14 in total) to fix all cables every 600 mm (2').
(Use one 813-929-00 in each position.)



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
9	655-181-03	CABLE UF2 SUPP BRD FO LONG W TAB OPTICAL FIBER	2

MACHINE:

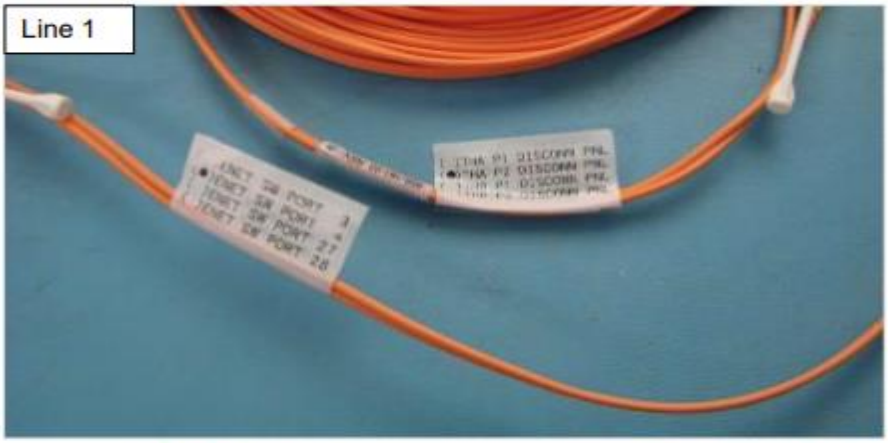
Marker

METHOD:

4.1 Take 2 UF2 SUPP BRD FO LONG w TAB optical fiber cables (655-181-03), and mark the optical fiber cables with labels "To" and "From" as shown in the figure.

①Mark “THA P1 DISCNN PNL” on one end and “ENET SW PORT 3” on the other end.

②Mark "THA P2 DISCNN PNL" on one end and "ENET SW PORT 4" on the other end.



Note: The optical fiber cable is easily damaged, so do not bend, twist, squeeze, throw or step on the optical fiber cable.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
10	655-185-03	CABLE UF2 INST FO TH TO SC W TAB OPTICAL FIBER	12

MACHINE:

Marker

METHOD:

5.1 Take 12 UF2 INST FO TH to SC w TAB optical fiber cables (655-185-03),
As shown in the figure, mark the 12 cables with the labels "To" and "From" of the optical fiber cables as follows:
One end is marked "THA P3-P12 DISCONN PNL", and the other end is marked "ENET SW PORT 13-24".

Note: The optical fiber cable is easily damaged, so do not bend, twist, squeeze, throw or step on the optical fiber cable.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
11	663-760-00	PLASTIC,DRAGON 12 SLOT FIBER OPTIC LABEL INFORMATION	1

MACHINE:

Tweezers

METHOD:

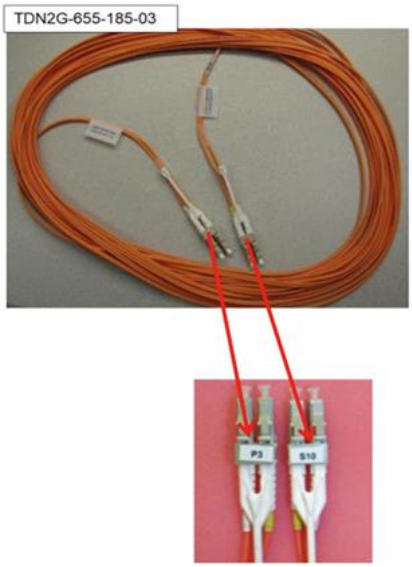
5.2 Take a label (663-760-00).

5.3 Tear off the "P3" and "S10" labels, and attach them to the two ends of one of the optical fiber cables 655-185-03 as follows. You can use tweezers to place the label on the connector surface.

5.4 Repeat step 5.3 according to the following table, and stick labels on both ends of the remaining 11 655-185-03 optical fiber cables.

Cable	Cable P/N	From TH Disconnect Panel	To TH Ethernet Switch in SC
1	655-185-03	P3	S10
2	655-185-03	P4	S8
3	655-185-03	P5	S6
4	655-185-03	P6	S4
5	655-185-03	P7	S2
6	655-185-03	P8	S0
7	655-185-03	P9	S1
8	655-185-03	P10	S3
9	655-185-03	P11	S5
10	655-185-03	P12	S7
11	655-185-03	P13	S9
12	655-185-03	P14	S11

TDN2G-663-760-00



Note: The optical fiber cable is easily damaged, so do not bend, twist, squeeze, throw or step on the optical fiber cable.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
12	634-324-04	PLASTIC,UF2 SPLITLOOM TH TO SC .75, HOSE,PIPE OR TUB	1

MACHINE:

Duct aid

METHOD:

6.1 Install all the optical fiber cables of the previous two steps into a gray sleeve (634-324-04).

A.655-181-03 UF2 SUPP BRD FO LONG w TAB (2 pcs)

B.655-185-03 UF2 INST FO TH to SC w TAB (12 pieces)

Tips: Use the duct aid as shown in the figure below to prevent damage to the optical cable!

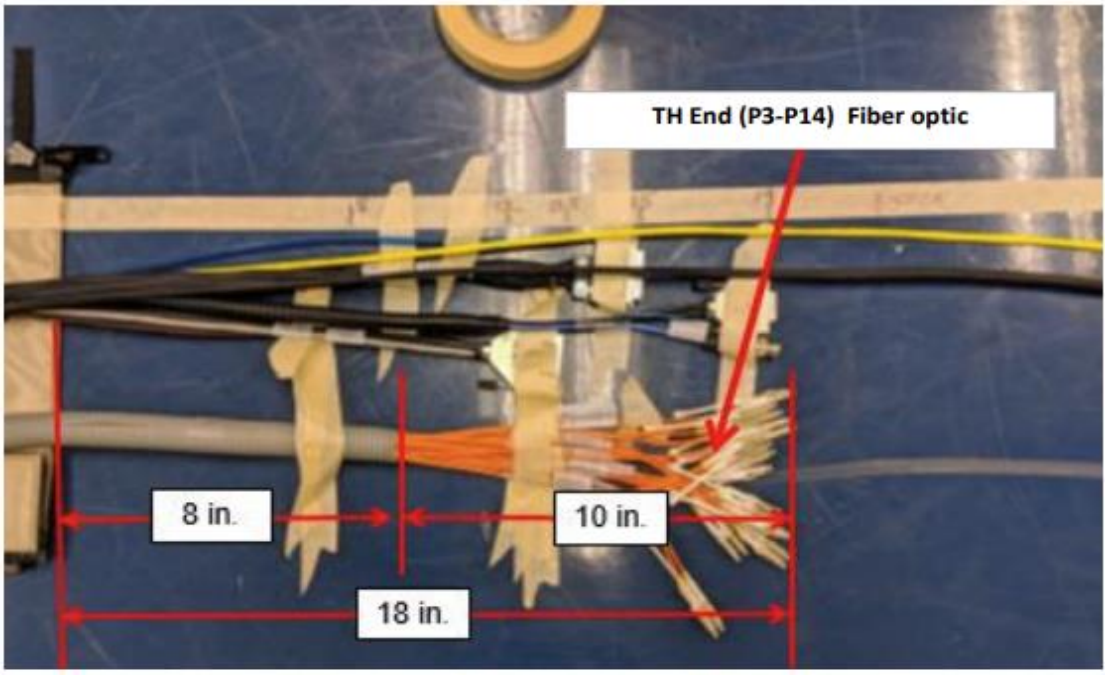
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

METHOD:

6.2 Ensure that the TH head end (P3-P14) of the optical fiber cable is installed at the front end of the cable sleeve/the TH head end, as shown in the figure below.
6.3 Place the gray fiber optic cable sleeve on the cable sleeve, and reserve 8 inches (203 mm) of the gray sleeve against the edge of the cable sleeve.
6.4 The optical fiber cable should be aligned with the edge of the gray sleeve, and the terminal of the optical fiber cable should extend 10 inches (254 mm) outside the gray sleeve.
This end is the "TH" end!



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
13	813-929-00	HDW VELCRO STRP 5/8	1

MACHINE:

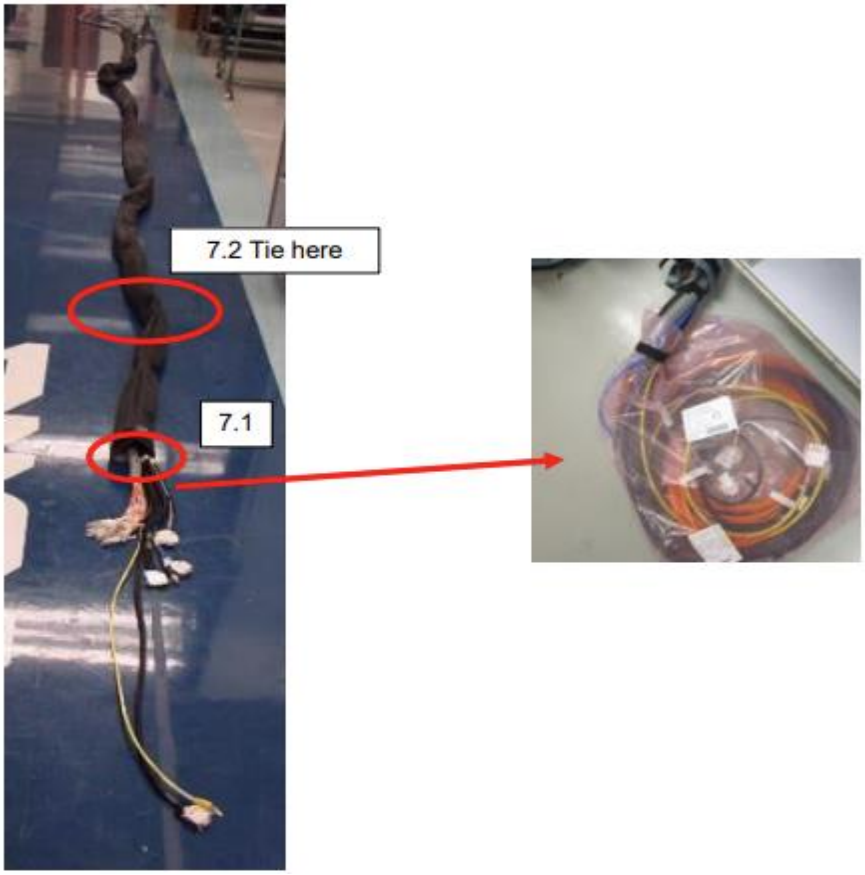
plastic bag

METHOD:

7.1 Use the existing cable tie to bundle the gray sleeve with all other cables.

7.2 Pull up the zipper of the thread sleeve, and then use a cable tie (813-929-00) to tie it to the edge of the thread sleeve to prevent the thread from slipping out of the thread sleeve.

7.3 Carefully roll up the excess cable at one end of the TH head and put it in a plastic bag for protection. Be extremely careful when handling fiber optic cables.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

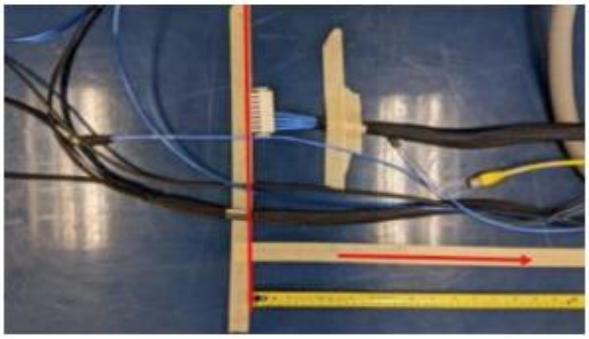
ITEM	P/N	DESCRIPTION	QTY
14	651-816-04	CABLE UF2 12SLOT SB25 SMCPWR INPUT TELECOM WITH	1
15	813-929-00	HDW VELCRO STRP 5/8	1

MACHINE:

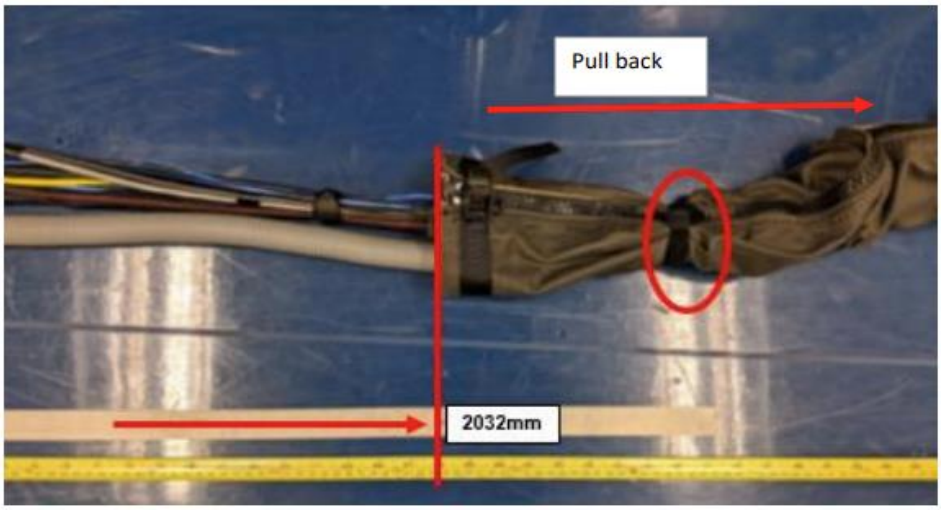
Measuring tape

METHOD:

7.4 Fix the 651-816-04 SMC_POWER cable connector on the floor with tape on the Cabinet end (rear) of the wire sleeve. Use a tape measure to measure a distance of 2032mm (80 ") from the end of the connector to the end of the cable sleeve, and mark on the paper tape.



7.5 Without changing the TH end of the cable sleeve, pull the edge of the cable sleeve backward (creasing) until it measures 2032mm (80") from the end of the connector.
7.6 Then use a cable tie (813-929-00) to tie it 1 foot away from the cabinet end edge of the cable sleeve to prevent the cable from slipping out of the cable sleeve.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

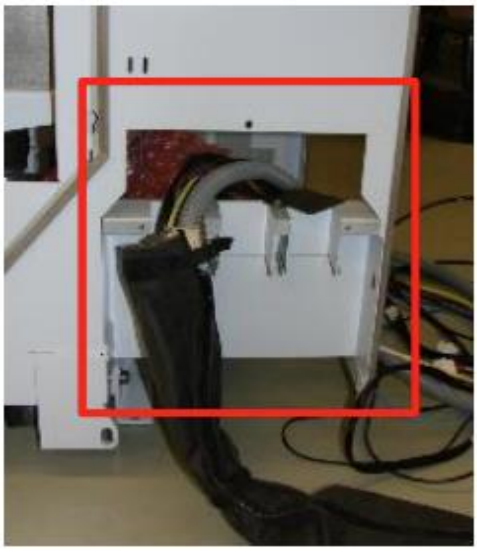
MATERIAL: N/A

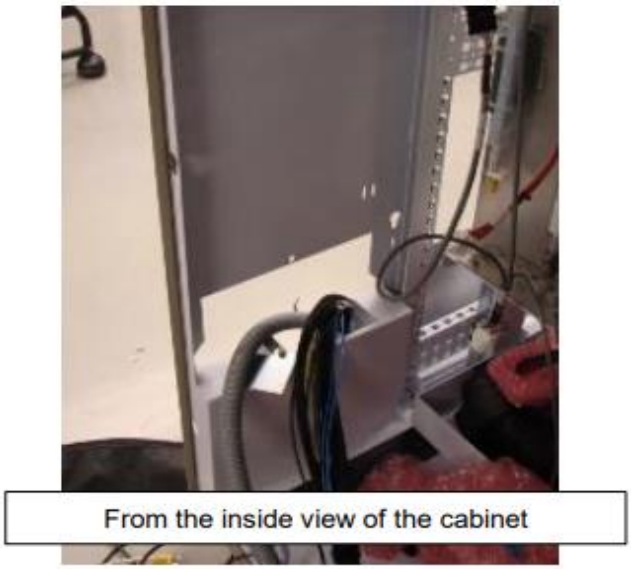
MACHINE: N/A

METHOD:

8.1 On the operating surface of the cabinet, near the lower left hand side of the cabinet, place the cable outlet position. Remove the cover and save 3 M5 screws for subsequent assembly.

8.2 As shown in the figure, unfold the cable at the end of the cabinet and install the cable into the cable outlet at the bottom of the bottom left corner of the operating surface of the cabinet. Put the thread into the passage on the left exit.





MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
16	813-929-00	HDW VELCRO STRP 5/8	3

MACHINE:

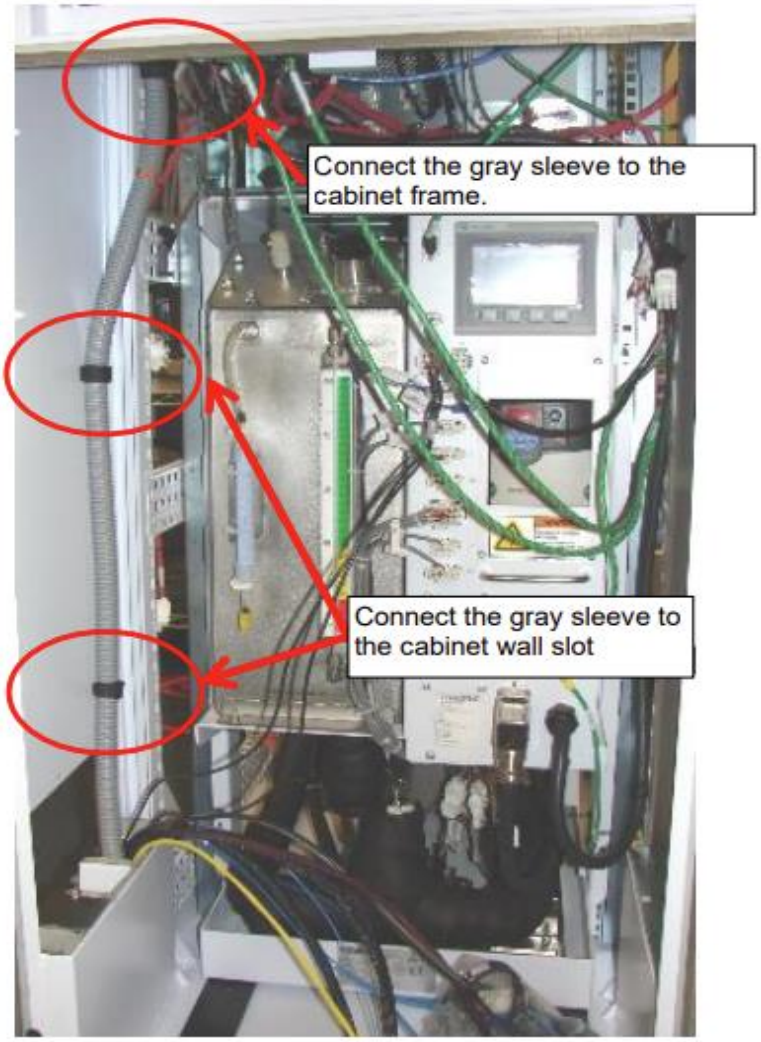
N/A

METHOD:

8.3 As shown in the figure, route the gray sleeve containing 14 optical cables upwards along the operating surface of the cabinet along the left side of the left cabinet frame.

8.4 Use three 813-929-00 cable ties to connect the casing to the cabinet side wall groove and frame at the position shown.

Note: The optical fiber cable is easily damaged, so do not bend, twist, squeeze, throw or step on the optical fiber cable.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
17	813-929-00	HDW VELCRO STRP 5/8	1

MACHINE:

N/A

METHOD:

8.7 As shown in the figure, use a cable tie (813-929-00) to roll up the excess tube/fiber cable and tie it to the inside of the cabinet top plate to keep the fiber end hanging.

Note: The optical fiber cable is easily damaged, so do not bend, twist, squeeze, throw or step on the optical fiber cable.

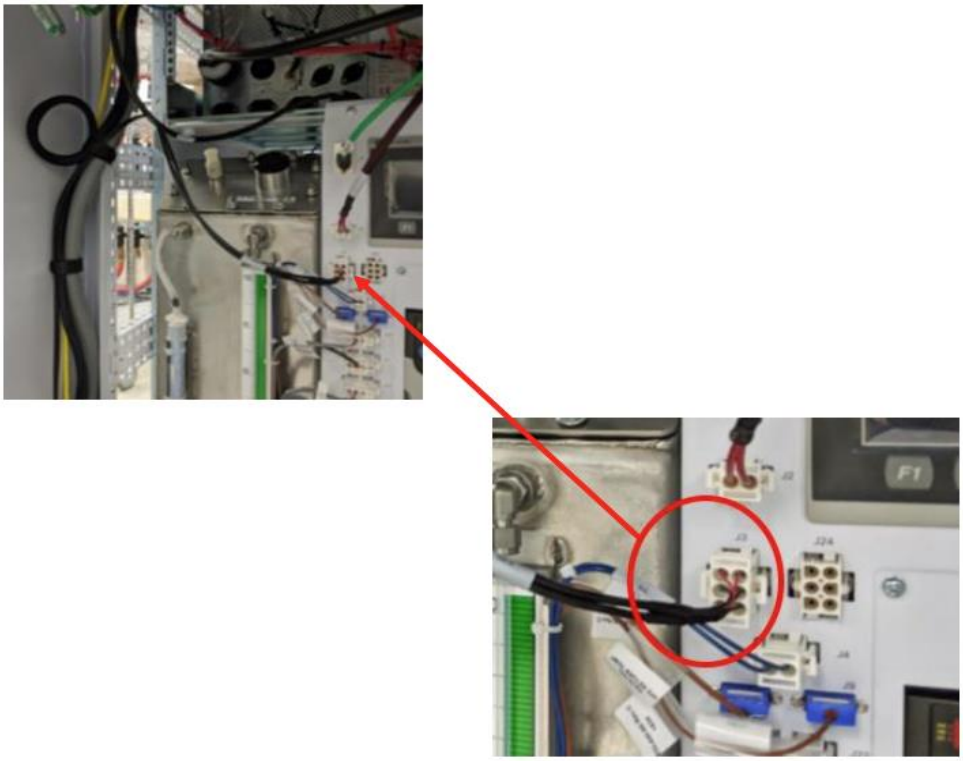
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

METHOD:

8.5 As shown in the figure, mark the terminal as "CDU J3" and route the cable along the gray sleeve to the CDU (E-box front panel), and insert it into the CDU J3 interface.
8.6 Secure the wire with the existing cable tie.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

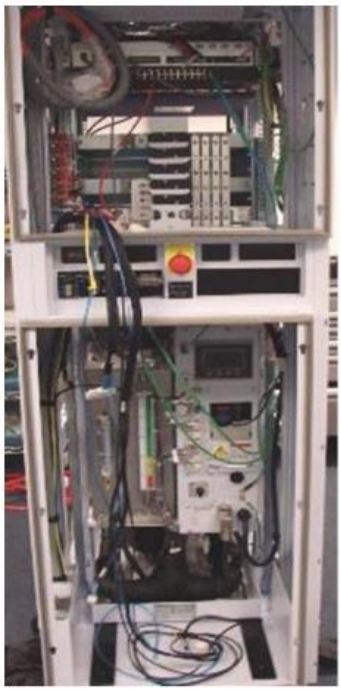
METHOD:

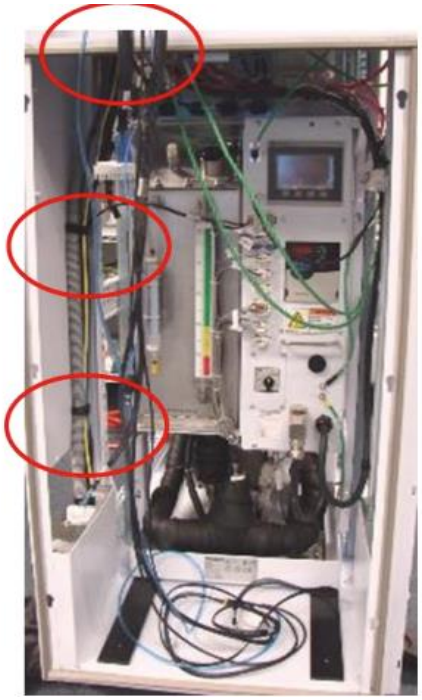
9.1 Route the cables listed below along the left side of the cabinet frame in front of Manifold upwards as shown in the figure. Pass the excess cables through the cabinet cover and temporarily hang them on the interface panel of the operation surface.

① 651-816-04-UF2 SMC Power Tray
② 651-111-02-LCL Enet To Support Cabinet
③ 657-472-03-UF2 SC TH SMCLON
④ 655-233-01-Dragon LCL External
⑤ 660-546-00-M6 air hose

Note: Put the 655-540-03-UF2 SC EMO LED cable under the power rack!

9.2 Use the existing cable tie to tie the cable to the EMO cable.





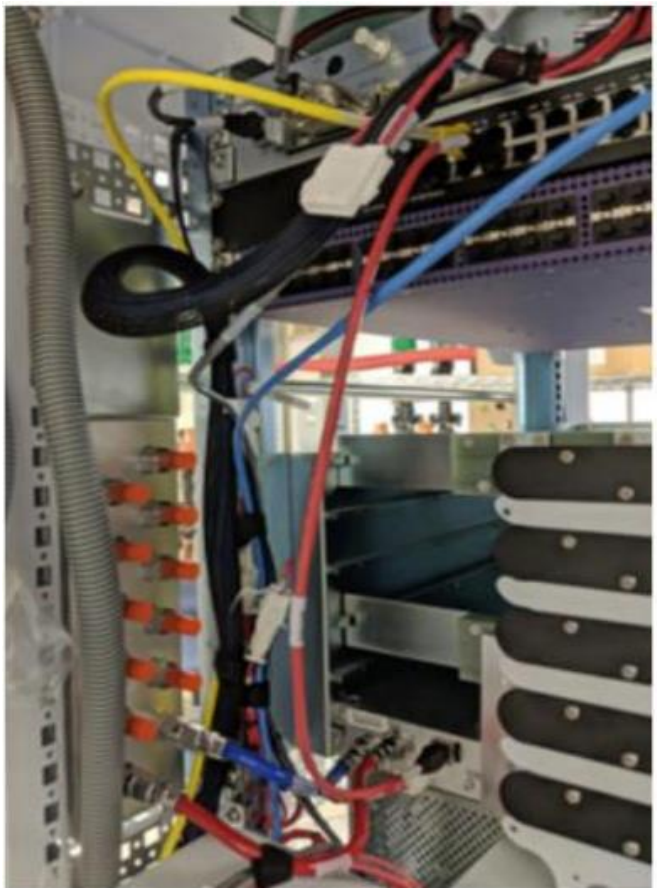
MAN: :CERTIFIED OPERATOR, TECHNICIAN

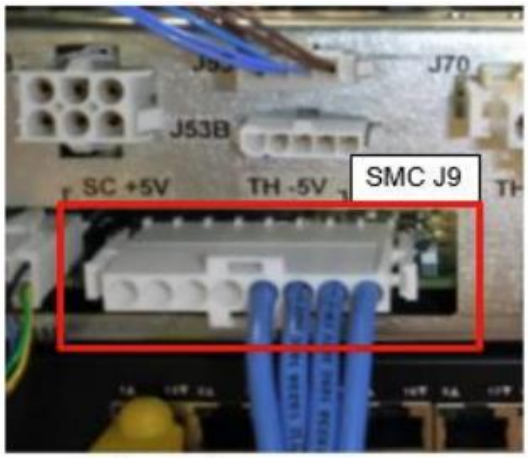
MATERIAL: **N/A**

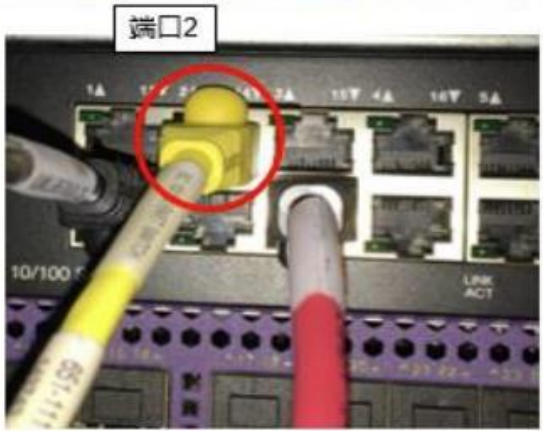
MACHINE: **N/A**

METHOD:

10.1 651-816-04-UF2 SMC power cord
a. As shown in the figure, place the adjacent optical fiber cable upward on the left side of the cabinet and place the connector of the optical fiber cable at the front of the SMC, and insert the connector into "SCM J9".
10.12 651-111-02-LCL Enet To Support Cabinet cable (yellow Cat 5 cable).
a. Connect the yellow cable installed on the left side of the cabinet to the Ethernet switch and plug it into port 2.







MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE: **N/A**

METHOD:

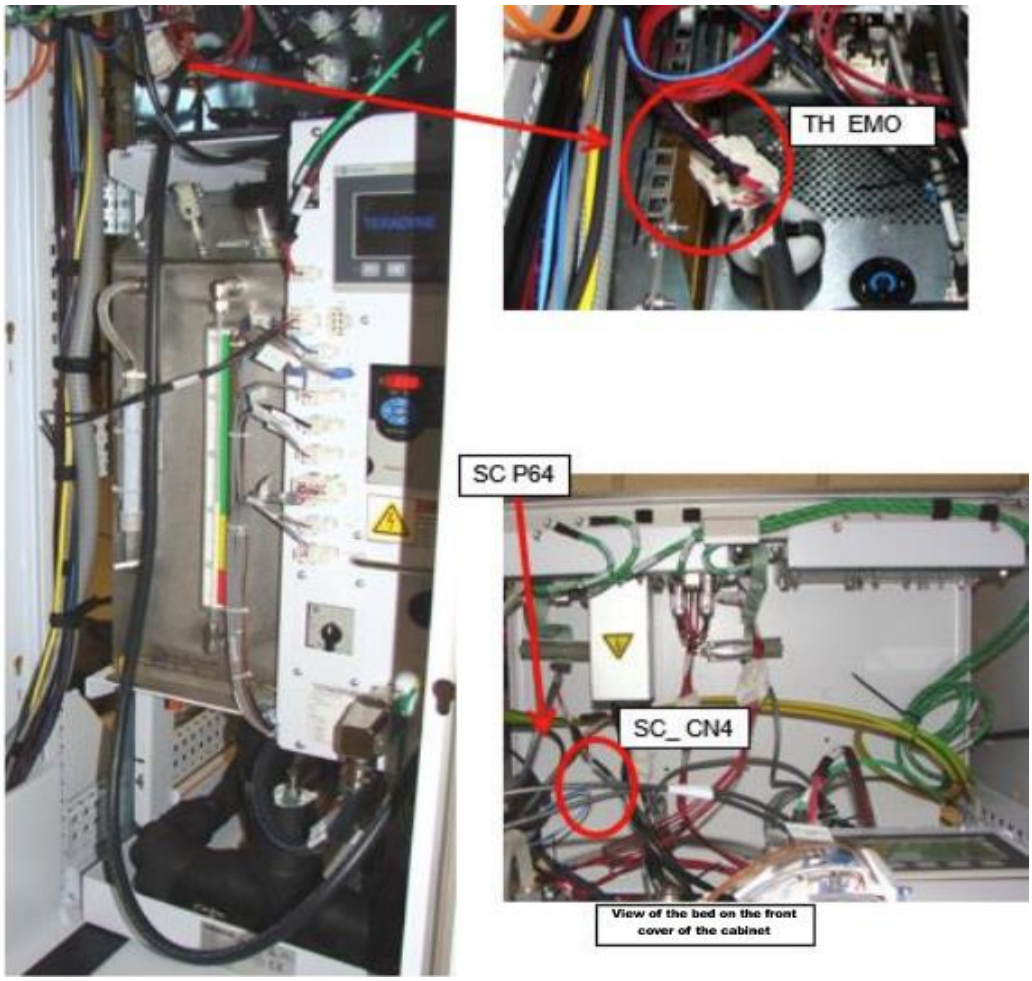
11.1 655-540-03-UF2 SC TH EMO LED

a. As shown in the figure, place the cable upwards along the installed gray sleeve on the left side of the cabinet

b. Connect the wire labeled "SC_EMO_TH_CN7" under the SC cover to the "TH EMO" connector of the cabinet near ADB as shown in the figure.

Connect the wire labeled "SC_CN4" to the "SC_CN4" connector of the cabinet.

c. Connect the wire marked "SC_ON_SW_64" to P64 of the cabinet as shown in the figure.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
18	813-929-00	HDW VELCRO STRP 5/8	2

MACHINE:

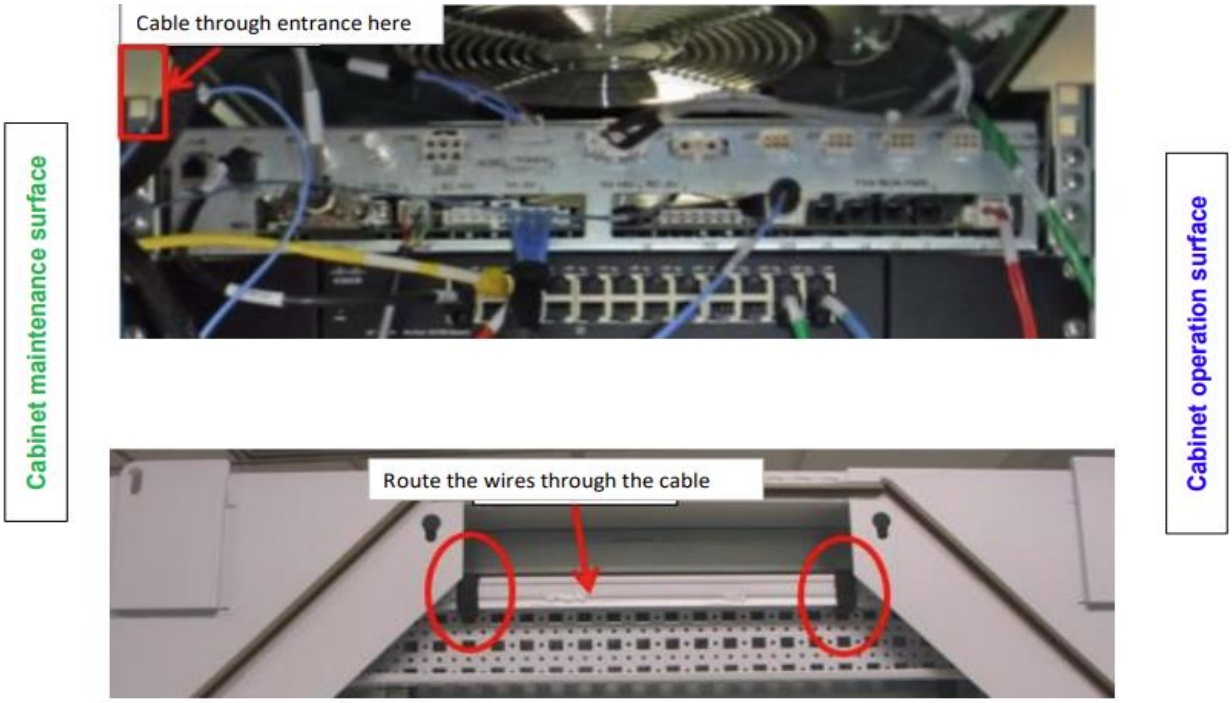
N/A

METHOD:

12.1 Route the following cables to the left side of the SMC tray as shown in the figure, and pass them through the slot entrance at the upper left corner of the cabinet to the maintenance side of the cabinet.

A. "SMC AV400 J5"
B.655-233-01 – Dragon LCL External;
C.660-546-00 – M6 air hose

12.2 Use 2 cable ties (813-929-00) as shown in the figure to tie the wire to the inside of the trough.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
19	813-929-00	Cable tie	1

MACHINE:

5/16 open torque wrench to tighten, 0.8 N·m (7 in-lbs)

METHOD:

12.3 Insert the "SMC AV400 J5" cable into the SMC AV400 "J5" interface.
Use the tool 5/16 open torque wrench to tighten, the torque value is 7 in-lbs.
12.4 Use a cable tie (813-929-00) to tie the SMC wire to the frame as shown in the figure.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

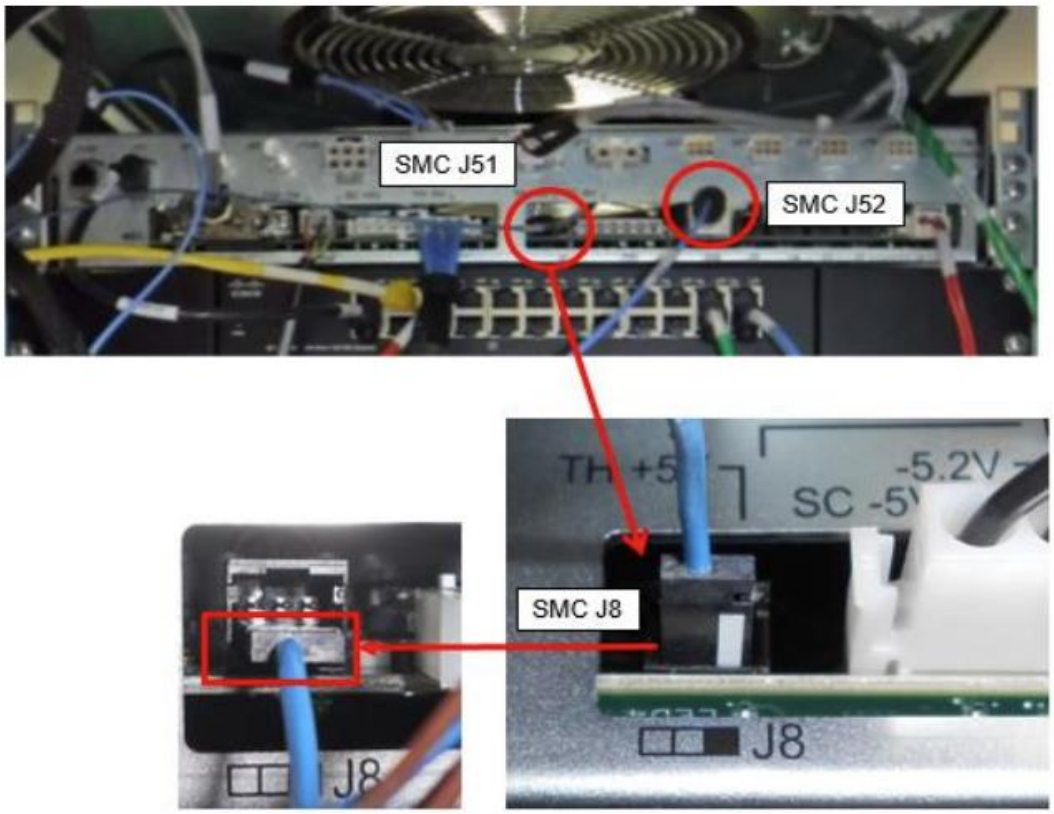
13.1 657-472-03-UF2 SC TH SMCLON

a. As shown in the figure, place the wire upwards along the installed wire harness on the left side of the cabinet

b. Install the connector marked "J51" into "SMC J51".

c. Mark "SMC-J8A" on the line and place it on the front of the SMC.

Insert the wire into the lower part of the connector as shown in the figure to ensure that the white stripes on the wire correspond to the black blocks on the silk screen



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

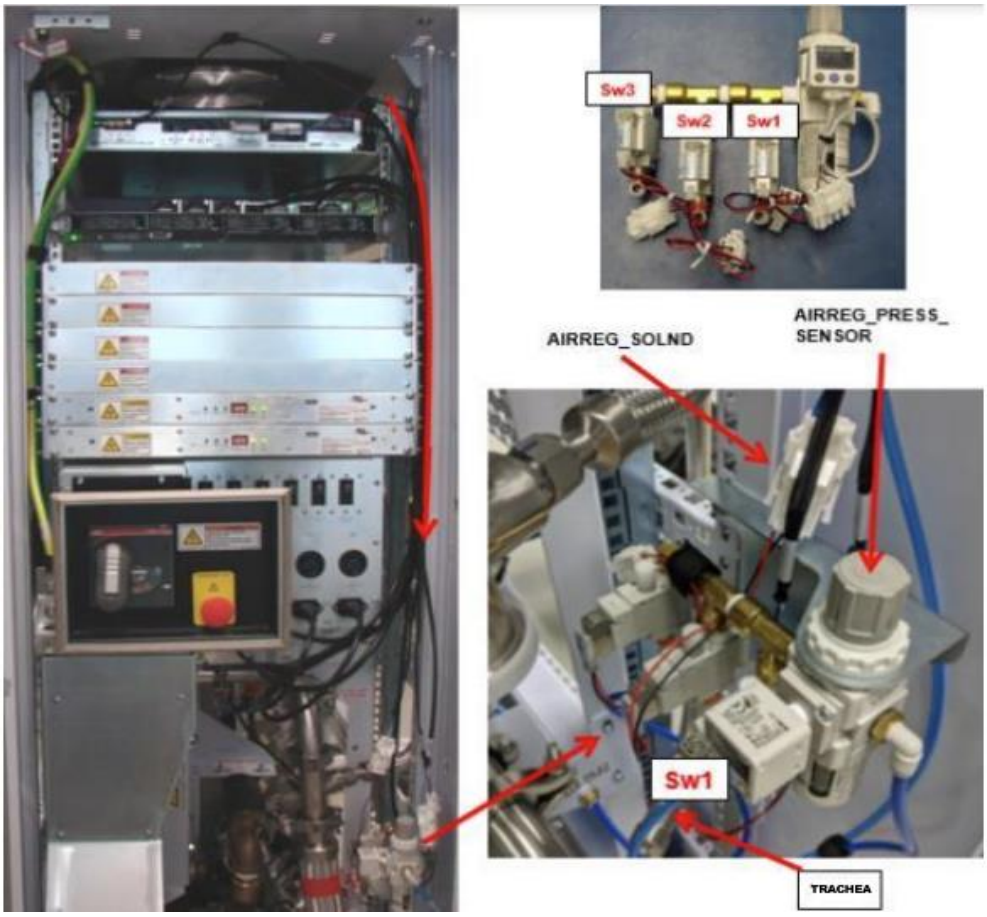
ITEM	P/N	DESCRIPTION	QTY
20	813-929-00	HDW VELCRO STRP 5/8	2

MACHINE:

N/A

METHOD:

14.1 Use two cable ties 813-929-00 to bind the LCL cable and 6mm hose, and tie them to the rack rail.
14.2 Connect the LCL cable end marked "SC_AIRREG_PRESS_SENSOR" to the air regulator pressure sensor connector.
14.3 Connect the LCL cable end marked "SC_AIRREG_SOLND" to the Sw1 connector.
14.4 Take a 6mm air pipe and connect it to the SW1 port of the air regulator as shown in the figure.
Note: Do not connect to Sw3 at this time!



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

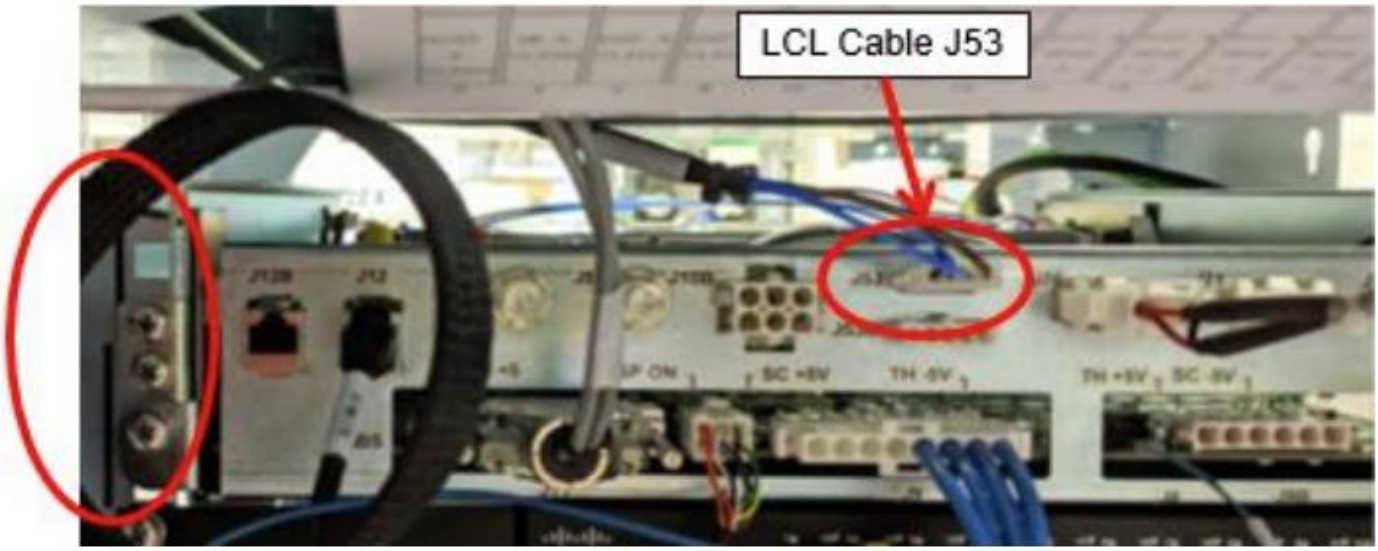
ITEM	P/N	DESCRIPTION	QTY
21	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE:

N/A

METHOD:

15.1 Route the "SMC J53" side of the cable 655-233-01 to the front of the SMC, and insert J53.
15.2 As shown in the figure, bind the cables on the left side of the cabinet and fix them with a cable tie (813-929-01).



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
22	710-048-10	TRANCEIVERS PIN SFP+ MULTI 10GBS HOT PLUG DUPLEX	1

MACHINE:

N/A

METHOD:

16.1 Find the following two connectors from the 655-181-03 UF2 SUPP BRD FO LONG w TAB optical fiber harness.

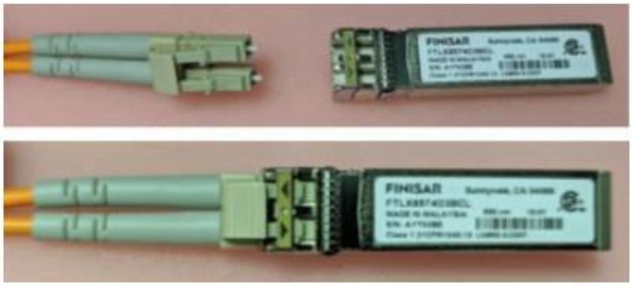
A. ENET SW PORT 3

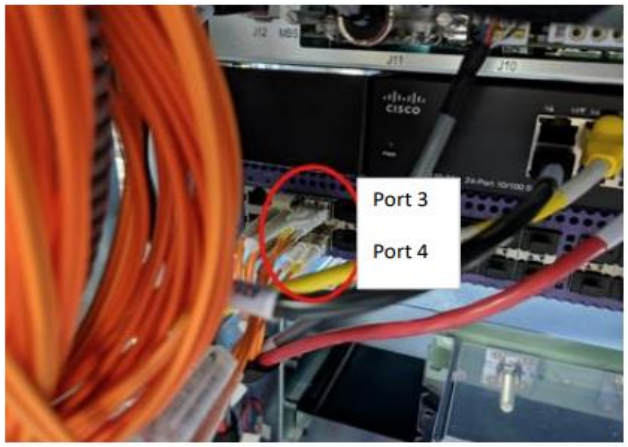
B. ENET SW PORT 4

16.2 Install 1 10G transceiver (710-048-10) on each of the two connectors (2 in total).

Remove the two protective caps from the cable and insert the transceiver until it clicks.

16.3 According to the label, plug the two connectors of the installed transceiver into the Ethernet switch port.





MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
23	813-929-00	HDW VELCRO STRP 5/8	2

MACHINE:

N/A

METHOD:

17.1 Route the signal cable and place it on the left side of the top of the cabinet as shown in the figure.

a. Use 2 cable ties (813-929-00) as shown in the figure to tie the cable to the side of the cabinet rail.

b. Do not put the wire in the guide rail.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
24	813-929-00	HDW VELCRO STRP 5/8	3

MACHINE:

N/A

METHOD:

17.2 Use 3 cable ties (813-929-00) to tie the wire and the gray sleeve together as shown in the figure.

Do not put the wire in the rail!

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
25	650-824-00	CABLE TH SUPPLY 48V WITHOUT CONN	1
26	650-825-00	CABLE TH RETURN 48V WITHOUT CONN	1
27	651-109-04	CABLE T/H 12SLT GROUND WIRE TELECOM WITH CONN	1
28	651-112-03	CABLE,UF2 12SLOT GND SC TO TH TERMBLK,TELECOM,WI	1

MACHINE:

Marker

METHOD:

18.1 Mark the following cables:

A. 1 red 48V power supply line (650-824-00), marked "5" at both ends of the line

B. 1 black 48V feedback line (650-825-00), marked "5" at both ends of the line

C. 1 ground wire (651-109-04), marked "SC GND PLATE G10"

D. 1 ground wire (651-112-03), marked "SC GND PLATE G9"

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
29	651-219-12	CABLE SOCK UF2 12-SLOT EMI CBL CVR BLACK	1

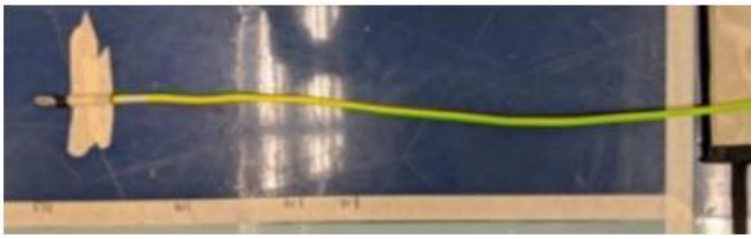
MACHINE:

FLX-CON-TAP-H00269 Masking tape

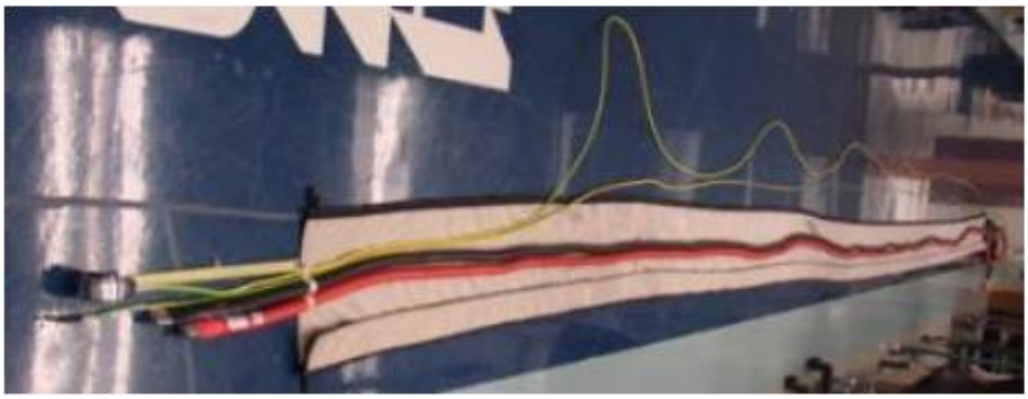
METHOD:

18.2 Place 1 EMI cable sleeve (651-219-12) on a flat surface and lay it in a straight line.

18.3 Stick a piece of masking tape on the floor, starting from the front edge of the thread sleeve.



18.4 Place the following thread on the thread sleeve.
① 650-824-00 red 48V Supply line-1 pc.
② 650-825-00 black 48V return line-1 piece.
③ 651-109-04 grounding wire-1 piece.
④ 651-112-03 grounding wire-1 piece.



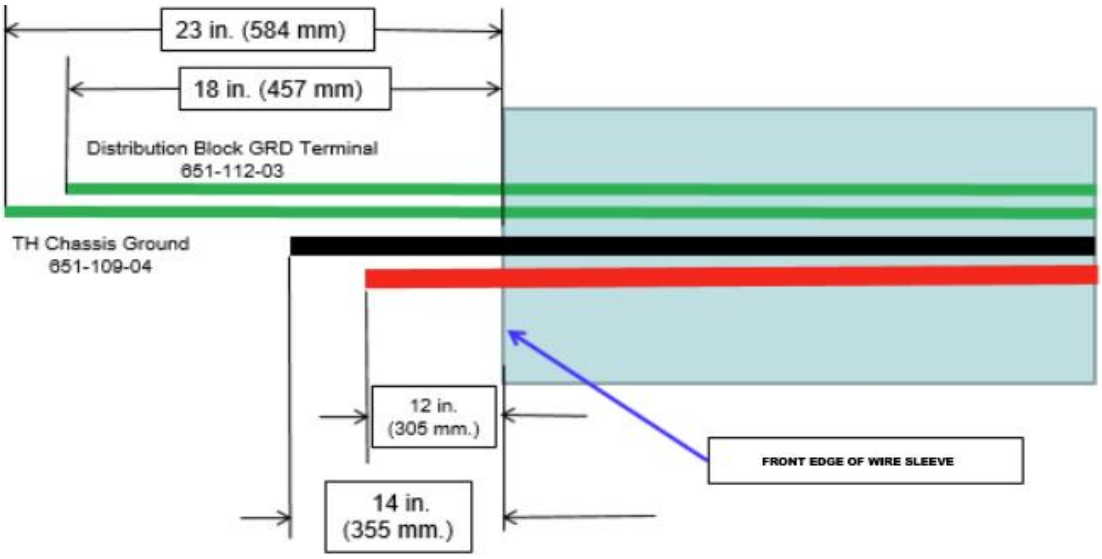
MAN: :CERTIFIED OPERATOR, TECHNICIAN

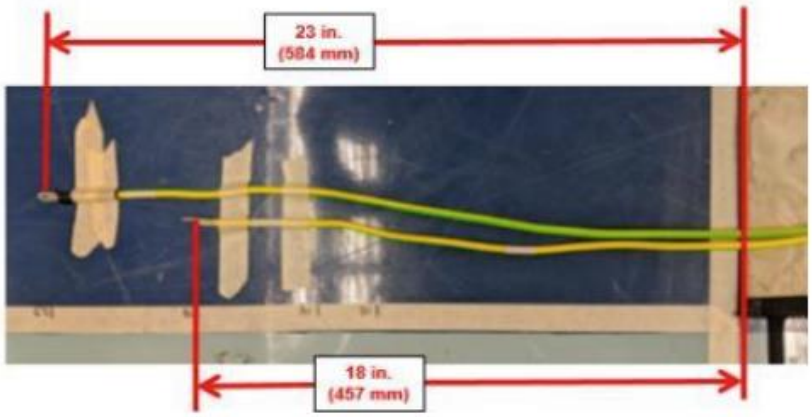
MATERIAL: N/A

MACHINE: N/A

METHOD:

18.5 The line should be aligned with the edge of the line sleeve to meet the following offset.
This will be the "test head" end.
Note: One end of the power cord with soft wire sleeve is at the cabinet end.





Note: The power cord is very heavy and requires two people to operate

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

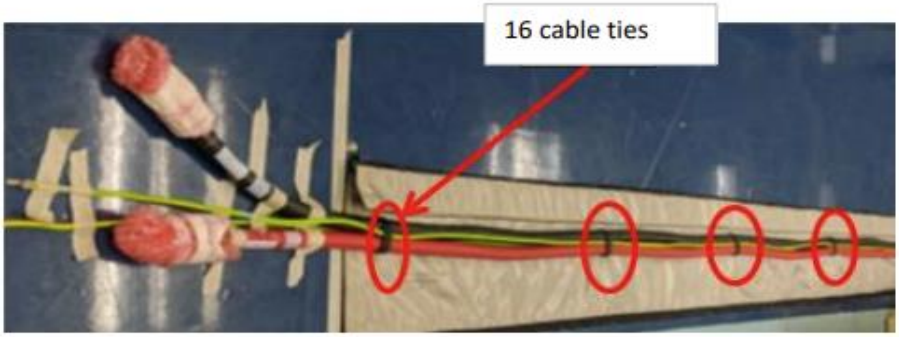
ITEM	P/N	DESCRIPTION	QTY
30	813-929-00	HDW VELCRO STRP 5/8	17

MACHINE:

N/A

METHOD:

18.6 Use 16 cable ties (813-929-00) to tie the wiring harness every 2 feet (610mm) as shown in the figure. Bundle the cables as shown in the figure below.



18.7 Close the zipper on the cable sleeve and tie a cable tie (813-929-00) about 8 inches from the end of the cable sleeve to keep the cable from sliding

Note: The power cord is very heavy and requires two people to operate.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

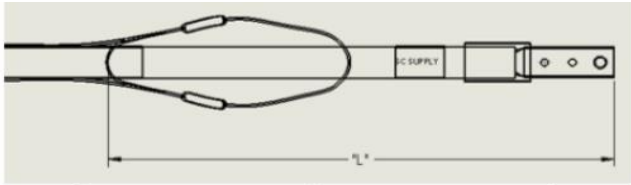
ITEM	P/N	DESCRIPTION	QTY
31	813-929-00	HDW VELCRO STRP 5/8	1

MACHINE:

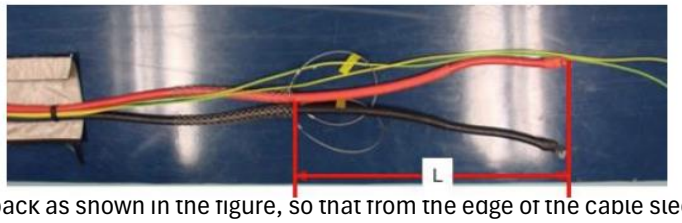
N/A

METHOD:

18.8 Adjust the position of the cord on the power cord-the wire at one end of the cabinet using the following table.

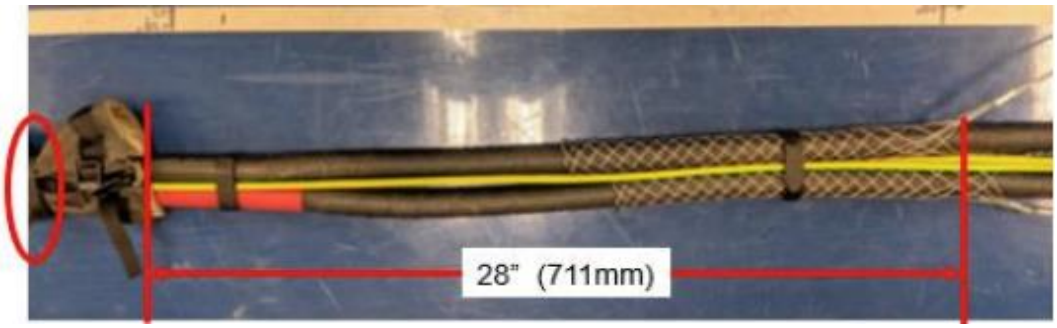


Cable Number	Length "L"
Red Cable	34 in (864 mm)
Black Cable	34 in. (864 mm)



18.9 Pull the cable sleeve back as shown in the figure, so that from the edge of the cable sleeve to the same rope position as before is 28 inches (711 mm).

18.10 Then use a cable tie (813-929-00) to fix the position on the wire sleeve.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

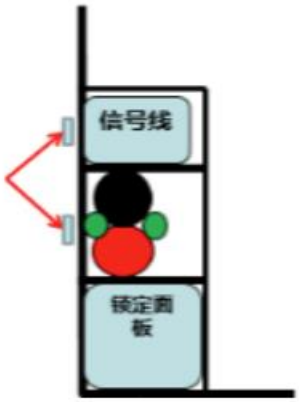
MACHINE: **N/A**

METHOD:

19.1 Install the main 48V power wire harness.

a. Install the power wiring harness into the middle channel at the bottom of the cabinet as shown in the figure.

b. Route the cables from the inside of the cabinet along the inner wall between the inner wall of the cabinet and the power rack. All(



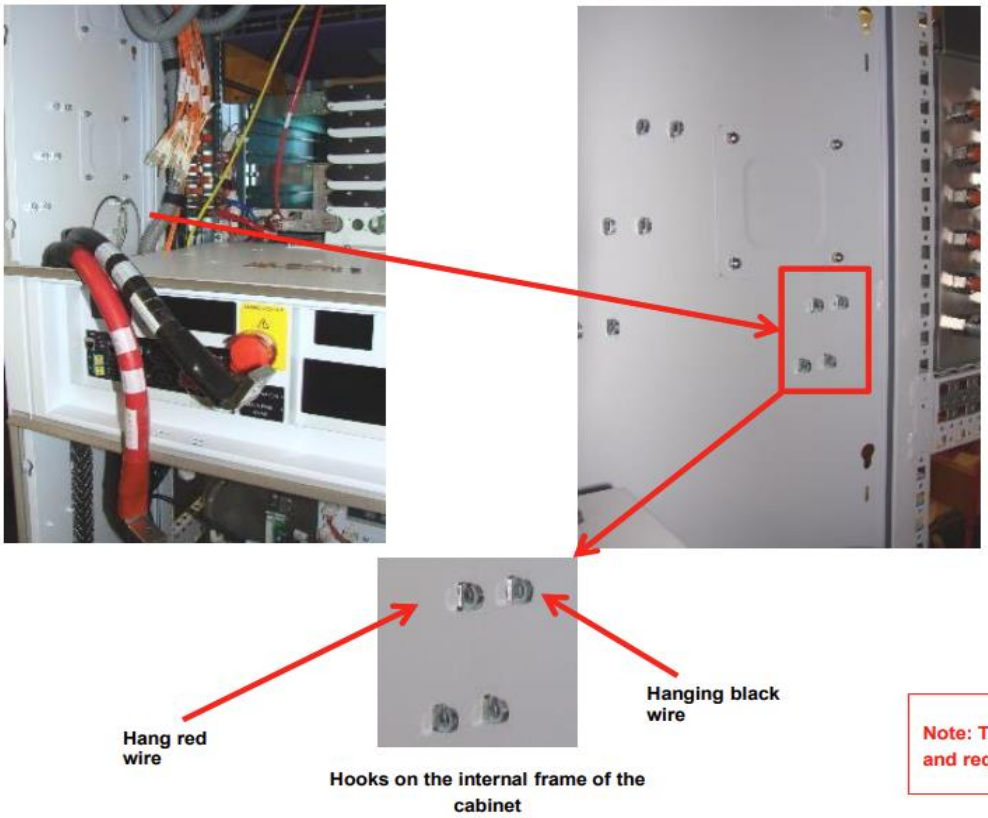
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

19.2 Hang the black and red cables on the inside of the cabinet frame as shown in the figure until the cables on the cables hang on the hooks on the cabinet.
Note that the color of the line corresponds to the position of the hook.



Note: The power cord is very heavy and requires two people to operate.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: 13mm socket wrench, 13.5 N·m

METHOD:

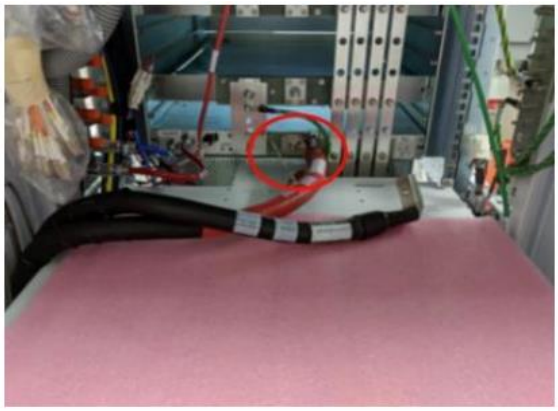
Note: Make sure that there is no heat shrink tube between the power cord and the PSU busbar.

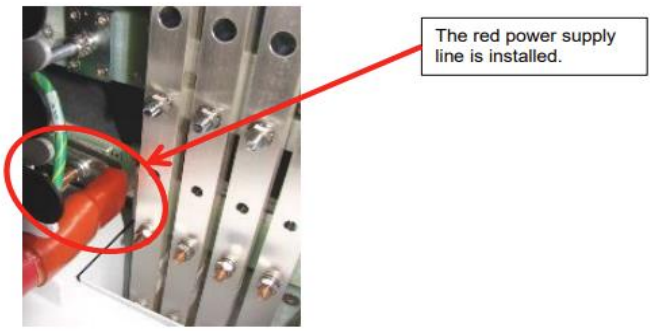
19.3 Temporarily place a pink ESD safety foam on the top of the power frame to avoid damage o the power frame during installation.

19.4 Remove the cable winding bracket from the power supply rack and save 5 M5 screws.

19.5 Install the red 48v power supply line (marked as 5) on the PSU power supply (+) terminal post, as shown in the figure below.

19.6 Use the original PSU M8 flat washer saved before, fix the M8 spring washer and M8 nut, the tool is a 13mm socket wrench, and the torque value is 13.5 N·m.





MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
32	603-096-01	NUT HEX SS M8X1.25 WSHR W/TTH 13MM WDH	1

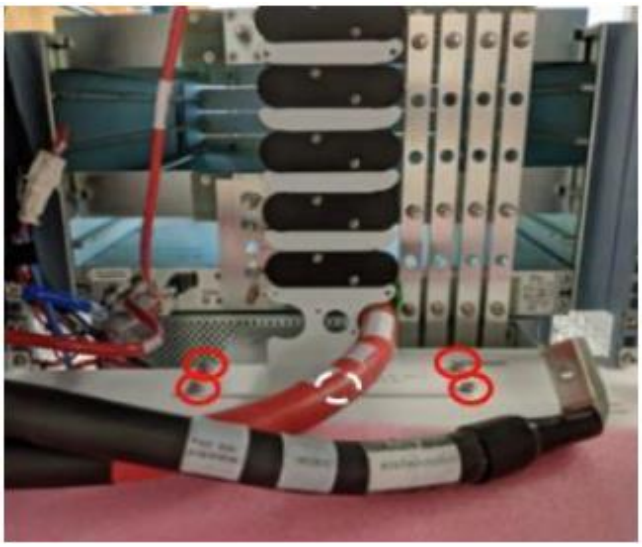
MACHINE:

13mm socket wrench 13.5 N·m, PH2 Phillips torque screwdriver 3.3 N·m

METHOD:

19.7 Install the "P1" end of line 658-670-00 on the PSU(+) stud with the nut just installed.
Use 1 M8 lock nut (603-096-01) to tighten, the tool is a 13mm socket wrench, and the torque value is 13.5 N·m.

19.8 Reinstall the previously removed cable winding bracket and fix it with the five saved M5 screws.
The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE:

13mm socket wrench, 13.5 N·m

METHOD:

Note: Make sure that there is no heat shrink tube between the power cord and the PSU busbar.

19.9 As shown in the figure, install the black 48V feedback cable (marked as 5) to the PSU feedback (-) terminal post on the left side of the power winding bracket,

19.10 Use the original PSU M8 flat washer saved before, fix the M8 spring washer and M8 nut, the tool is a 13mm socket wrench, and the torque value is 13.5 N·m.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE:

PH2 Phillips torque screwdriver, 3.3 N·m.

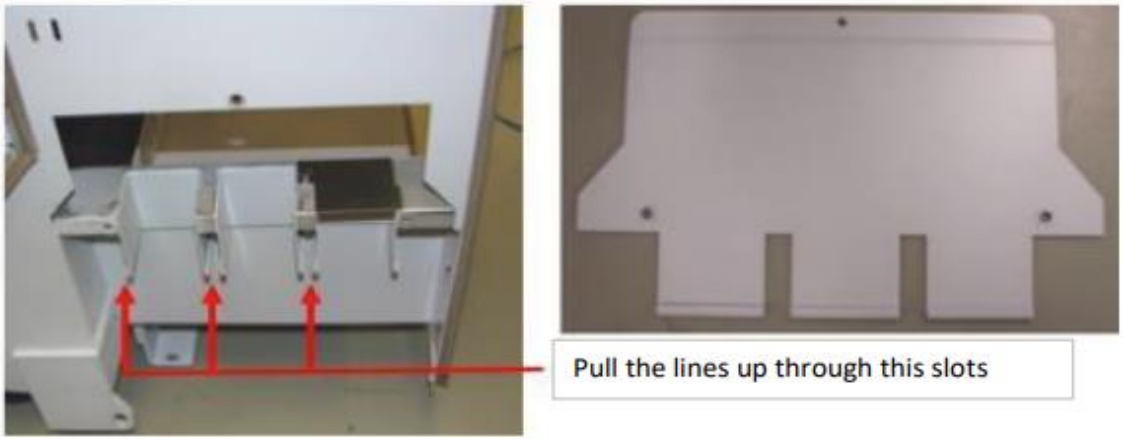
METHOD:

20.1 Install the EMI cable sleeve on the cabinet.

a. Try to put the wire vertically in the channel of the cabinet.

b. Insert the edge of the wire sleeve into the slot as shown in the figure, and pull up the zipper on the wire sleeve on the front and back. Make sure that the wire sleeve covers the edges of these slots.

c. Covers kept between installations.



d. Use the two M5 screws saved before to fix the cover on the cabinet, the tool is a PH2 Phillips torque screwdriver, and the torque value is 3.3 N·m.

e. Tighten the cord buckle on the cable sleeve as shown in the figure and tie it to the cabinet frame.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
33	813-929-00	HDW VELCRO STRP 5/8	2
34	813-929-01	HDW VELCRO STRP 5/8	3

MACHINE:

N/A

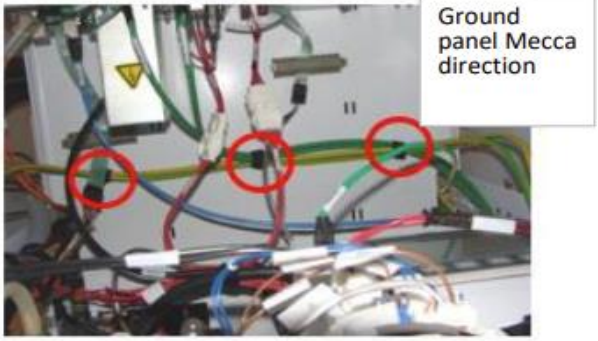
METHOD:

21.1 Grounding wire installation

a. Route the yellow-green ground wire from the left side of the cabinet along the 48V power supply line, as shown in the figure, use two (813-929-00) cable ties to tie the 48V power supply line.

b. Then lead the ground wire from under the power frame cover to the MECCA ground panel on the right side of the cabinet.

c. Use 3 (813-929-01) cable ties to bind the grounding wire to other installed grounding wires, and tie them to the power rack slot together.



Bottom overview

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

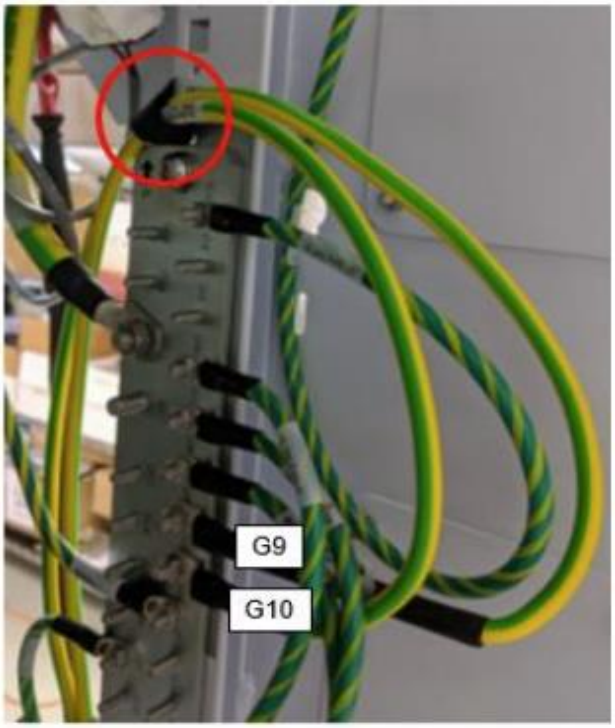
ITEM	P/N	DESCRIPTION	QTY
35	603-096-00	NUT HEX SS M5X0.8 W/WSHR W/TTH 8MM WDH	2
36	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE:

13mm socket wrench, 3.3 N·m.

METHOD:

d. Pass the ground wire through the top of the cabinet rail and place it on the MECCA ground panel.
Use two M5 nuts (603-096-00) to fix the ground wire.
① 651-109-04 SC GND PLATE G10 Connect to the G10 port of MECCA grounding panel,
the tool is a 13mm socket wrench, and the torque value is 3.3 N·m.
② 651-112-03 SC GND PLATE G9 is connected to the G9 port of MECCA ground panel,
the tool is a 13mm socket wrench, and the torque value is 3.3 N·m.
e. As shown in the figure, use a cable tie (813-929-01) to fix the ground wire to the cabinet.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
37	603-093-45	STANDOFF 10MM HEX M-F M5X.8X20MML SS	1
38	470-502-05	WASHER LK SPLT M5, 9.2MM OD, S	1

MACHINE:

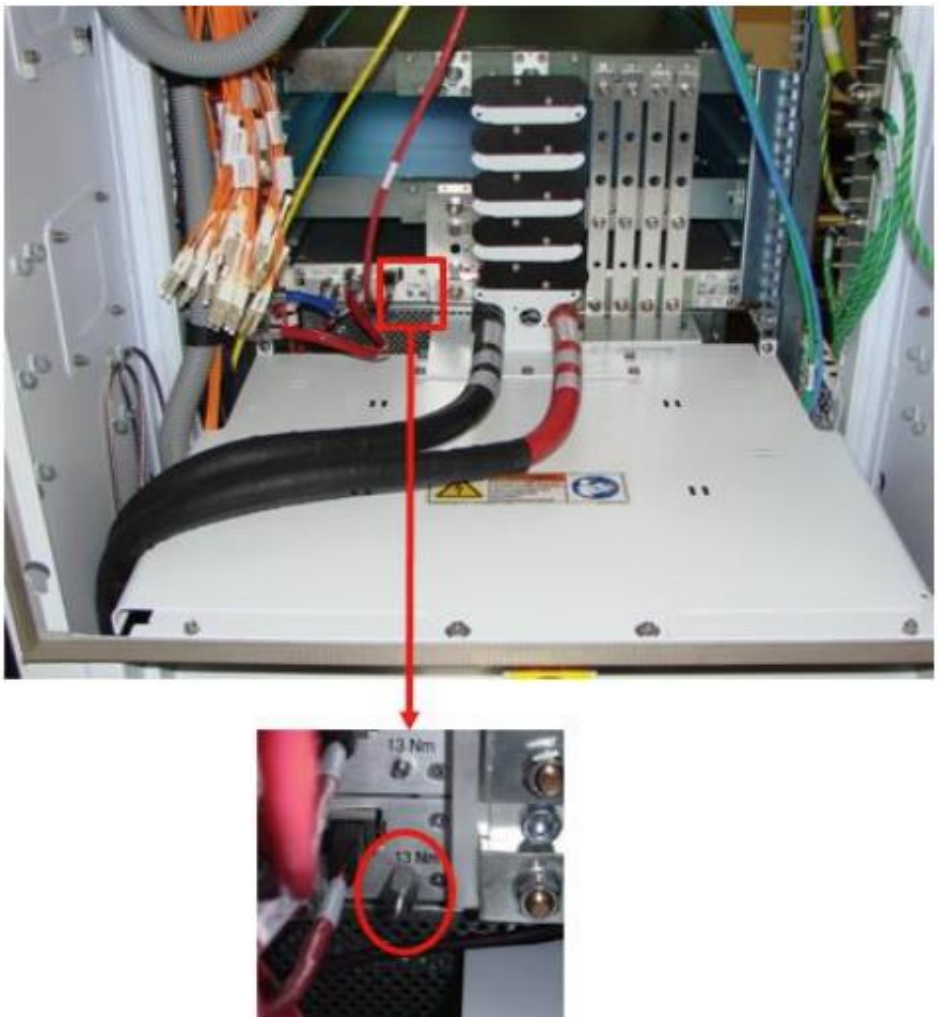
10mm socket wrench, 3.3 N·m

METHOD:

22.1 Installation of Busbar plexiglass protective cover

a. Install 1 (603-093-45) stud and 1 M5 spring washer (470-502-05) to the stud behind PSU 5.

The tool is a 10mm socket wrench with a torque value of 3.3 N·m



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
2	651-816-04	CABLE UF2 12SLOT SB25 SMCPWR INPUT TELECOM WITH	1
3	655-540-03	CABLE UF2_SC_TH_EMO_LED_INTLK_A SM CONN TELECOI	1
4	657-472-03	CABLE UF2 SC TH SMCLON_PRESS_DISCON_THA TELECOI	1
5	651-111-02	CABLE UF2 12SLT LCL ENET TH TO SC TELECOM WITH CON	1
6	655-233-01	CABLE UF2 12 SLOT LCL EXTERNAL CONTROL TELECOM W	1

MACHINE:

PH2 Phillips torque screwdriver 1.6 N·m, 8mm socket wrench 1.6 N·m.

METHOD:

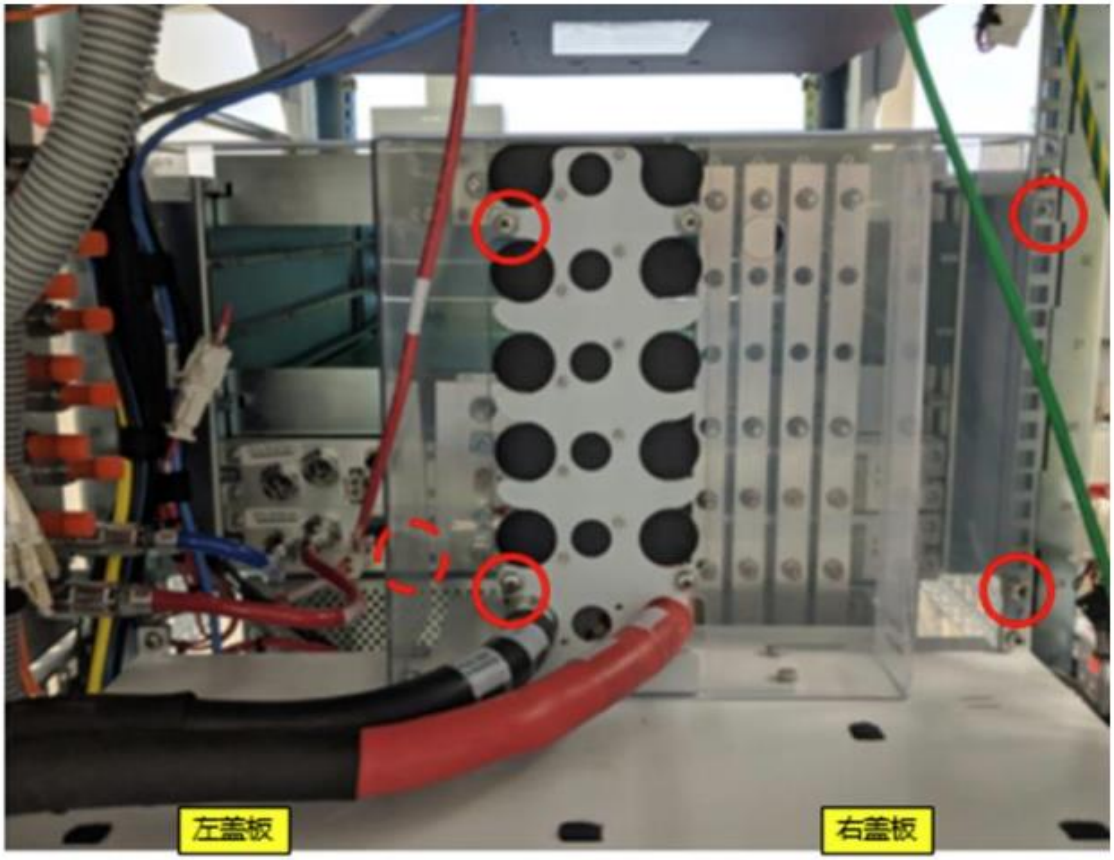
b. Take two protective covers and carefully remove the film on the protective covers:
①Left protective cover 660-866-50 1 piece
②The right protective cover 660-866-51 1 piece
c. Use 4 M5 screws (472-426-12) to fix the right protective cover 660-866-51 as shown in the figure,
The tool is a PH2 Phillips torque screwdriver with a torque value of 1.6 N·m.
d. First use 2 M5 screws (472-426-12) to fix the left protective cover 660-866-50 to the winding bracket as shown in the figure.
The tool is a PH2 Phillips torque screwdriver with a torque value of 1.6 N·m.
Use one M5 nut (603-096-00) to fix the left protective cover 660-866-50 to the stud installed before PSU5,
The tool is an 8mm socket wrench with a torque value of 1.6 N·m.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:



Note: Remove the film on the protective cover before installing the protective cover.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

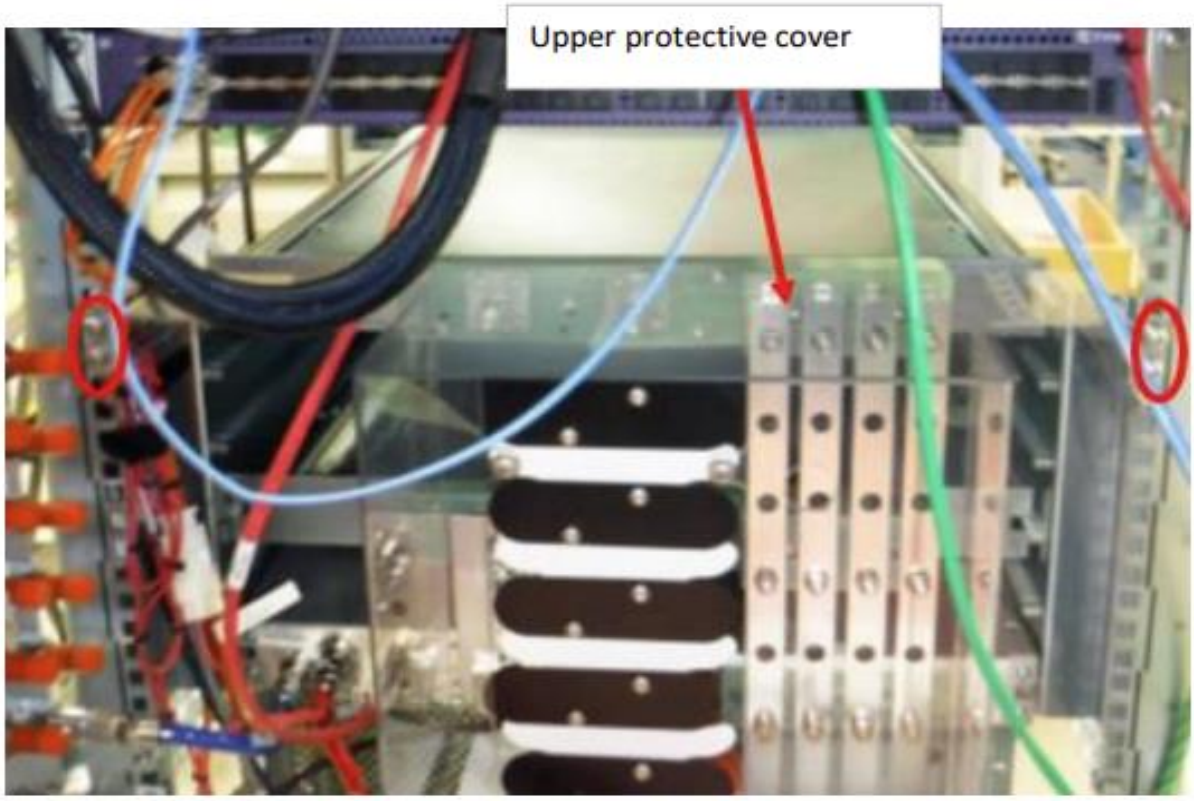
ITEM	P/N	DESCRIPTION	QTY
44	660-866-52	MECHANICAL,UF2 SC BUSBAR COVER,TOP,FOR SEMI TEST	1
45	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8x12MM L W/(	4

MACHINE:

PH2 Phillips torque screwdriver 1.6 N·m.

METHOD:

e. Take 1 (660-866-52) upper protective cover, and carefully remove the film on the protective cover:
f. Use 4 M5 screws (472-426-12) to fix the upper protective cover 660-866-52 as shown in the figure,
The tool is a PH2 Phillips torque screwdriver with a torque value of 1.6 N·m.



Note: Remove the film on the protective cover before installing the protective cover.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

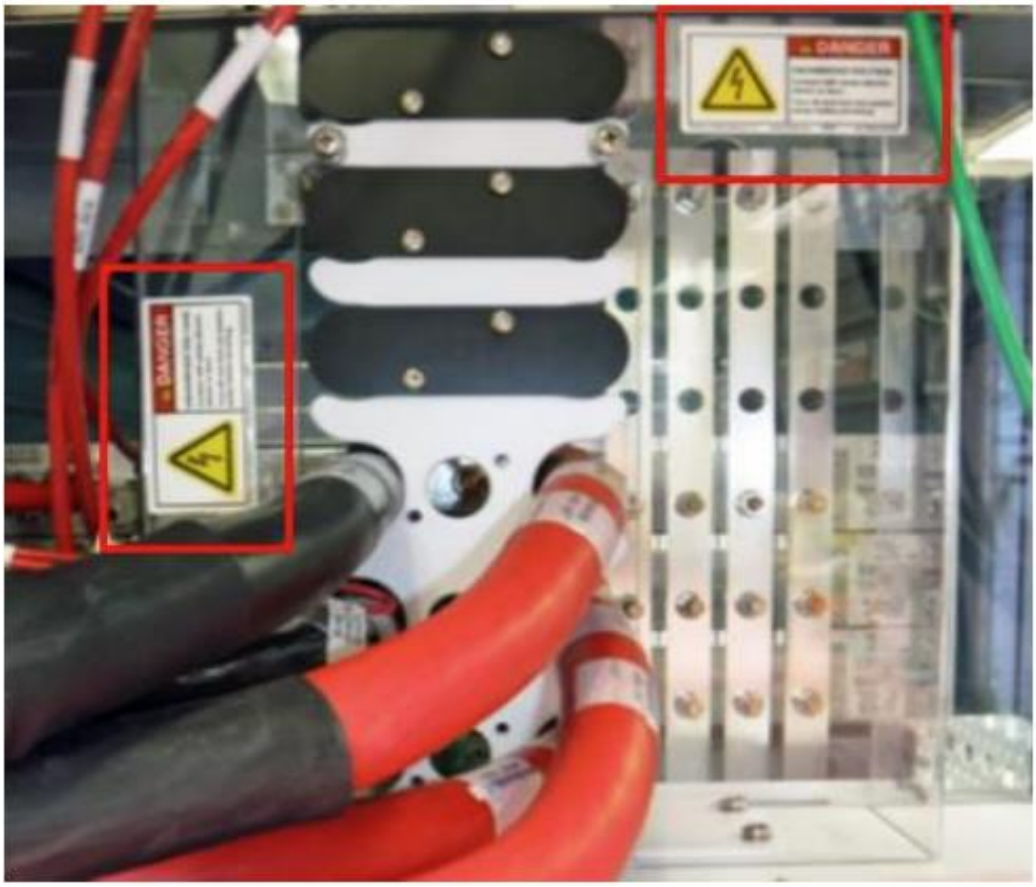
ITEM	P/N	DESCRIPTION	QTY
46	479-133-18	LBL DANGER HAZ VLT 2.75X1.35"	2

MACHINE:

N/A

METHOD:

g. Attach two danger labels (479-133-18) on the two plastic covers at the position shown in the figure-pay attention to the direction.
Right side cover: Align the upper edge of the label to the upper edge of the cover.
Make sure that the label does not extend beyond the top of the cover. Center the lid horizontally.
Left cover: Align the long side of the label to the left edge of the cover. Make sure that the label does not extend beyond the edge of the cover.Center vertically on the lid.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

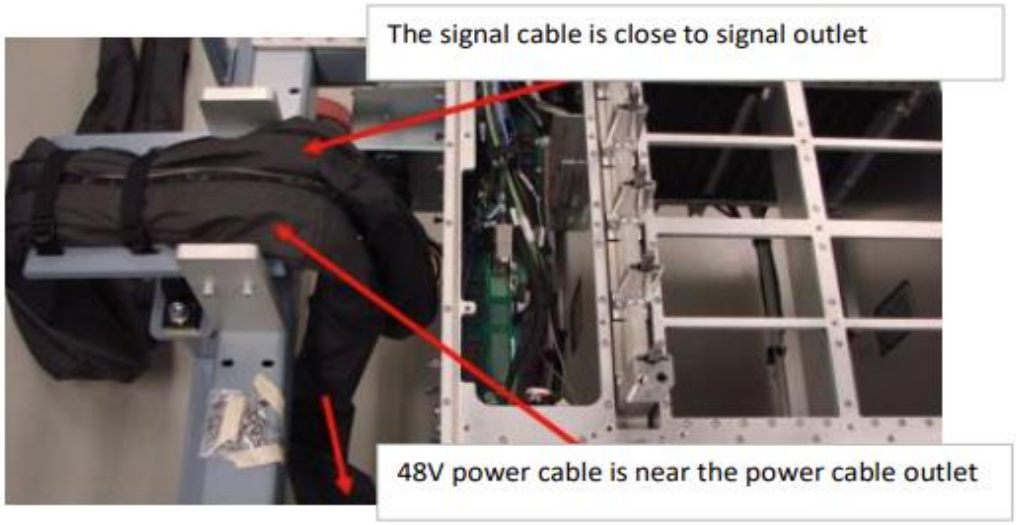
METHOD:

23.1 T/H to cabinet assembly-power cord assembly

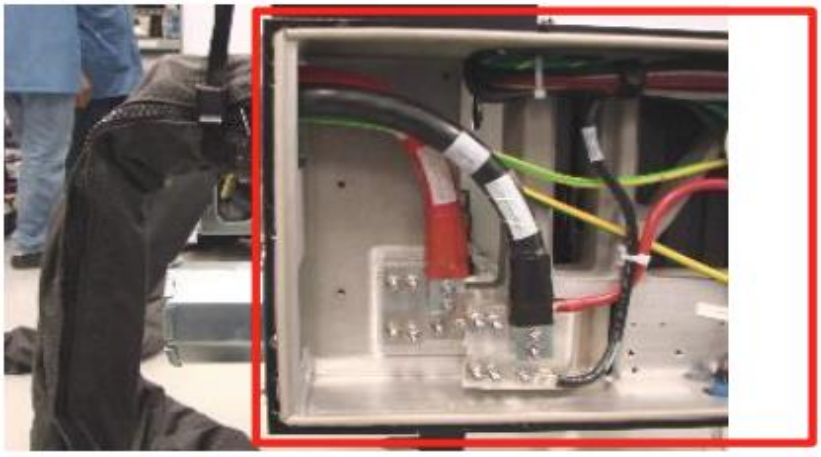
a. Rotate the T/H direction as shown in the figure, and route the signal wiring harness to the top of the beam as shown in the figure;

b. As shown in the figure, pass the 48V power supply wire harness through the top of the beam and face the lower cable outlet.

c. Fix the two wire harnesses on the rack with nylon straps.



d. Rotate the T/H direction 90° as shown in the figure, and the T/H power supply part faces upward.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE:

10mm socket wrench 5.6 N·m

METHOD:

e. Insert the 48V power supply harness into the T/H power slot through the top cable outlet (closest to the T/H center line).

①Place the 48V power supply line in the busbar position as shown in the figure-plug he red line plug into the busbar as shown in the figure.

② Make sure that the black power cord is placed crosswise on the red power cord.

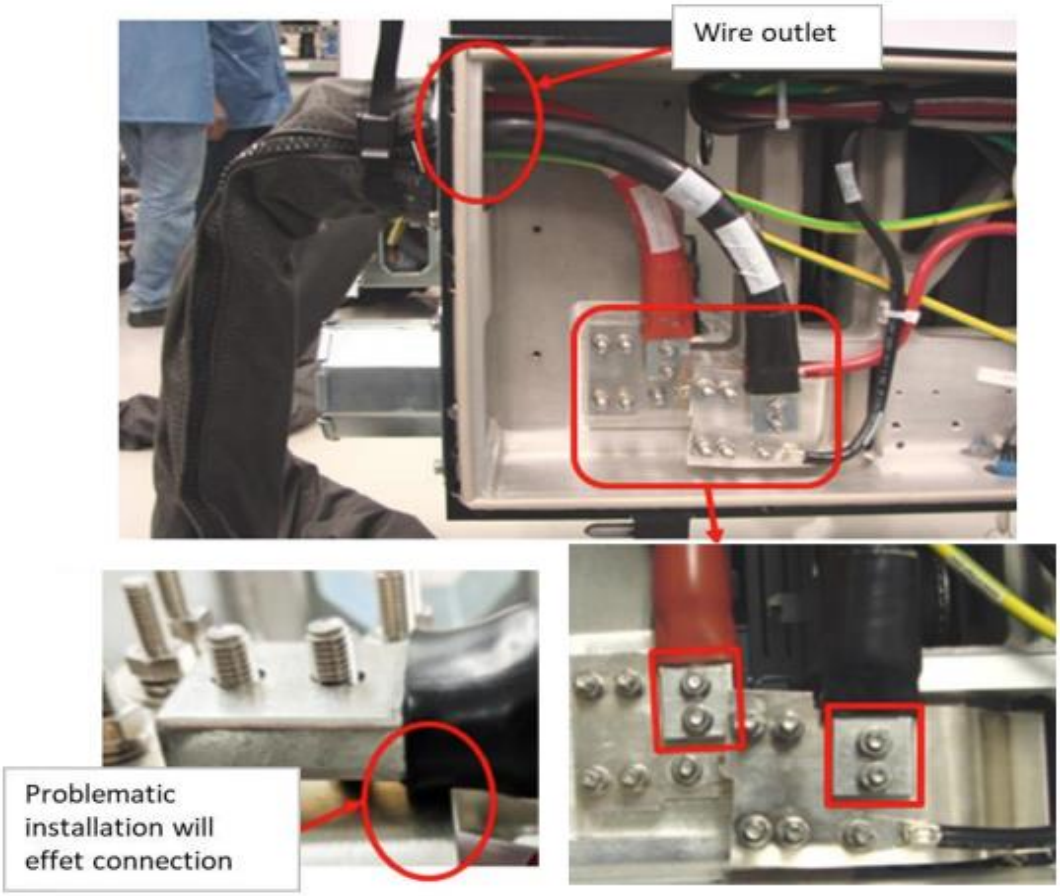
Warning: Ensure that the insulation of the power supply line does not press on the Busbar.

If necessary, the insulating material can be trimmed so that it will not be touched during installation.

③Use the 4 sets of M6 flat washers and M6 nuts that the Busbar comes with to fix the power cord plug on the busbar in 4 positions.

The tool is a 10mm socket wrench with a torque value of 5.6 N·m

④Tighten the strap of the cable cover and fix the cable cover on the cable outlet.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

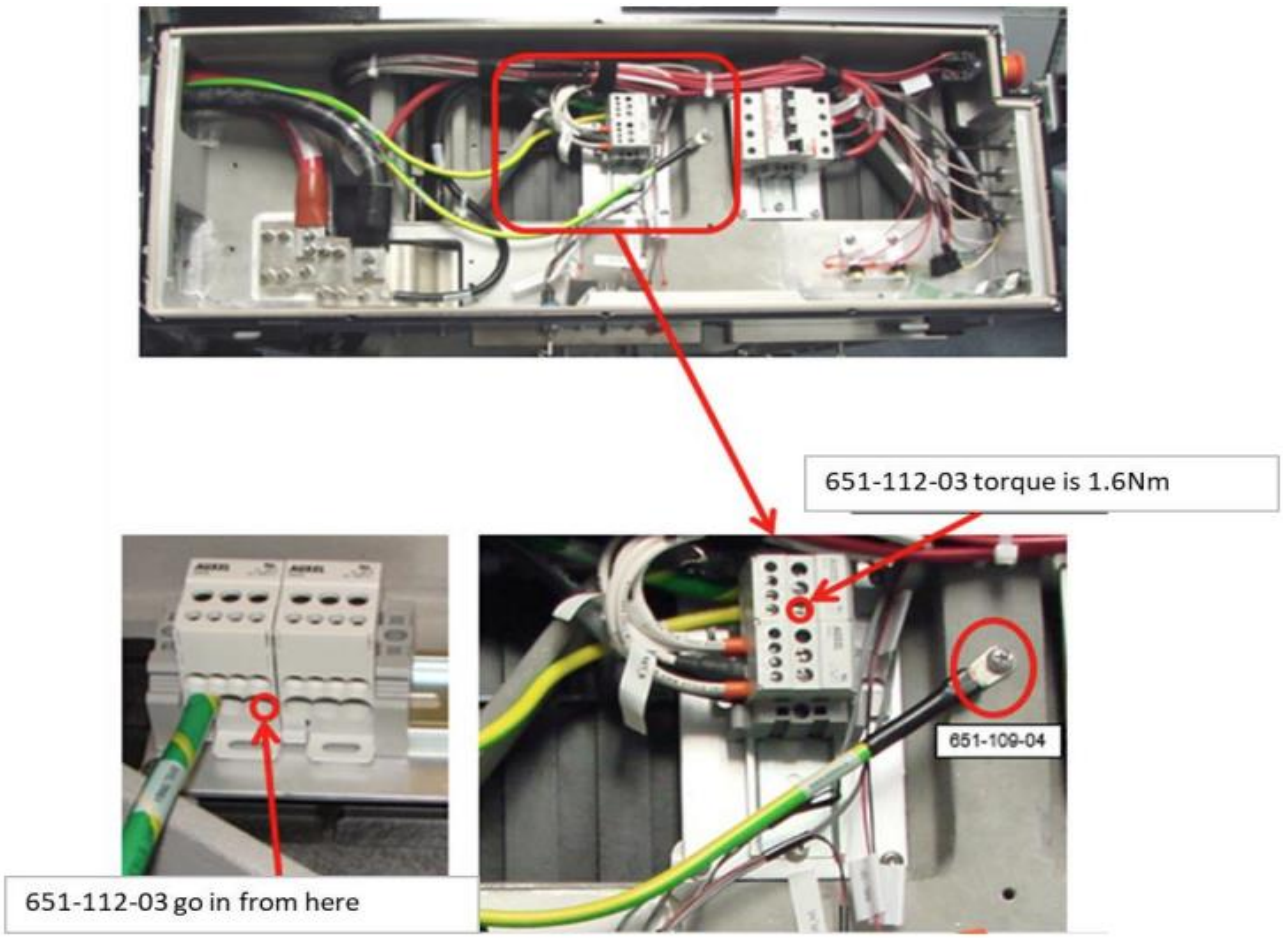
ITEM	P/N	DESCRIPTION	QTY
47	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8x12MM L W/(	1

MACHINE:

PH2 Phillips torque screwdriver 1.6 N·m, PH2 Phillips torque screwdriver 3.3 N·m

METHOD:

f. Connect the two ground wires to the power distribution block
and the circuit breaker as shown in the figure.
① Install the ground wire 651-112-03 to the power distribution terminal block.
Use the tool PH2 Phillips torque screwdriver, the torque value is 1.6 N·m.
② The ground wire 651-109-04 is fixed to the chassis with one M5 screw (472-426-12).
Tools PH2 Phillips torque screwdriver, The torque value is 3.3 N·m.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

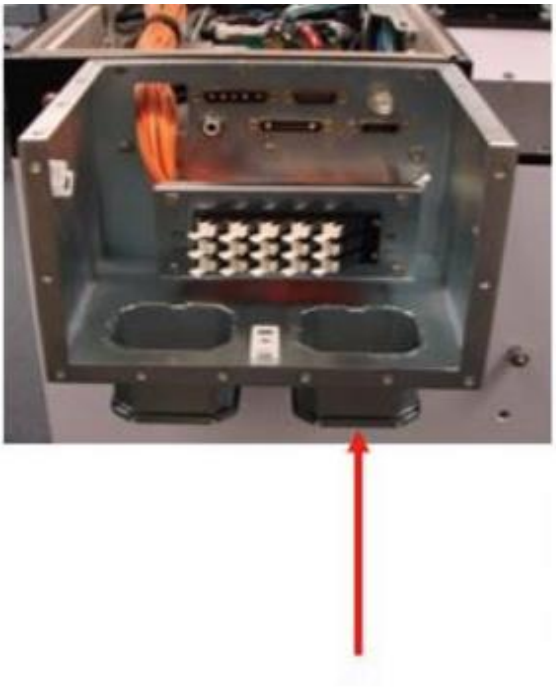
METHOD:

24.1 T/H to cabinet assembly-signal line assembly

a. Observe the circuit breaker panel from the rear of the T/H as shown in the figure, and insert the signal wire harness upward through the wire outlet on the right.

b. Insert the gray sleeve with the optical fiber cable and the remaining signal wires through the right wire outlet.

c. Insert and drop the two wires 631-554-00 and 633-460-00 from the wire outlet on the left. These two wires will be connected to the PTL at the back.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

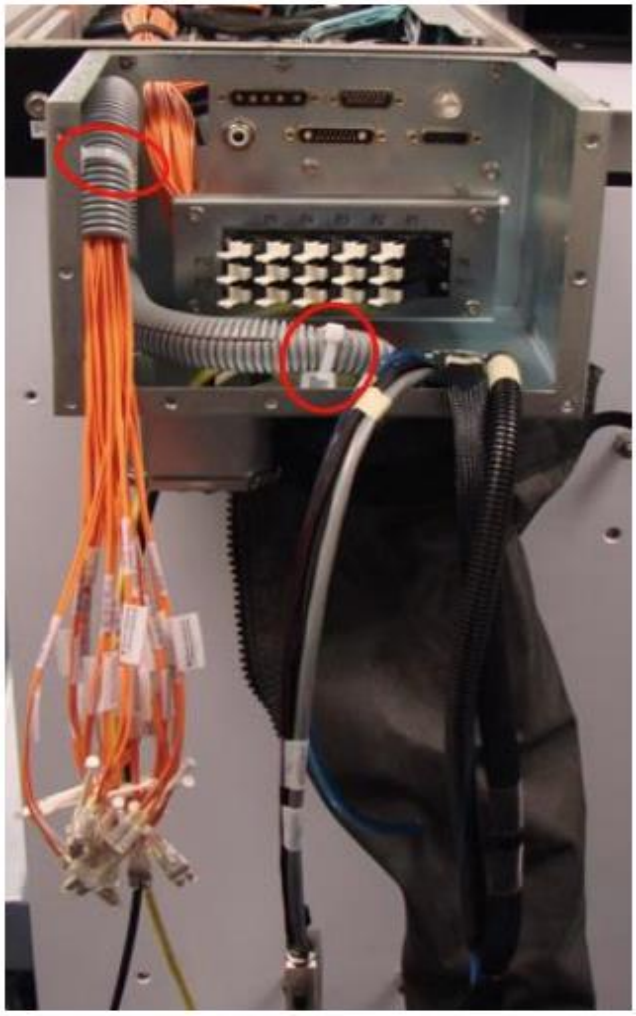
ITEM	P/N	DESCRIPTION	QTY
48	938-102-00	TIE WRAP 6.5 X 3/16"	2

MACHINE:

Cutter

METHOD:

d. As shown in the figure, roll the orange optical fiber harness (gray tube) into a 3/4 turn and tie it to the clamp with 2 cable ties (938-102-00).



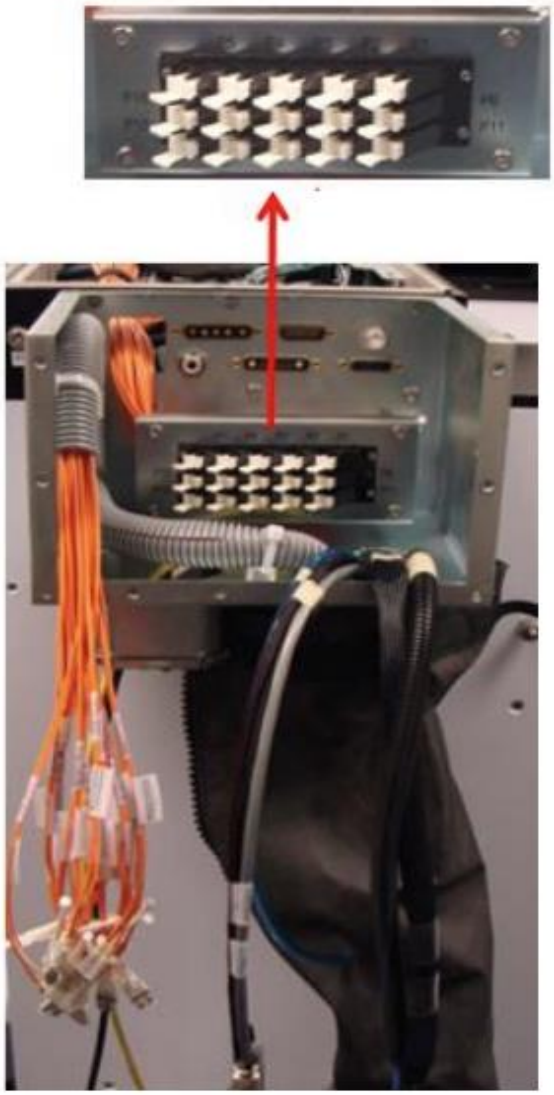
MAN: :CERTIFIED OPERATOR, TECHNICIAN

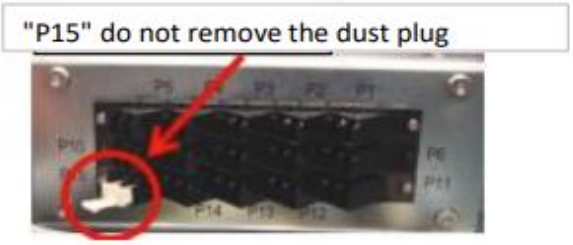
MATERIAL: N/A

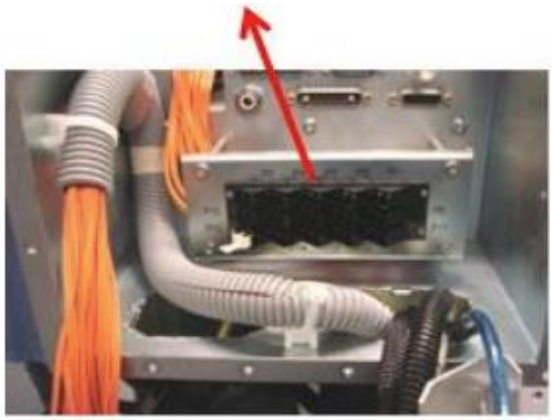
MACHINE: N/A

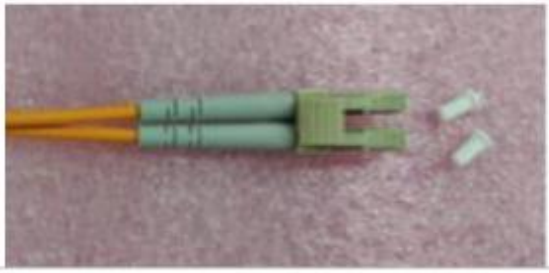
METHOD:

e. Remove the dust plugs at all positions except "P15" from the optical interface panel of the disconnect panel as shown in the figure.
save these dust-proof plugs and reinstall them when button up.









Remove the dust plug. Save it to be re-install at Button Up.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

g. Insert the optical fiber cable into the port on the adapter panel as shown in the right figure on the left table.

Group 1 (1st row)					
	P15	P14	P13	P12	P11
CABLE P/N	OPEN	655-185-03	655-185-03	655-185-03	655-185-03

Group 2 (2nd row)					
	P10	P9	P8	P7	P6
CABLE P/N	655-185-03	655-185-03	655-185-03	655-185-03	655-185-03

Group 3 (3rd row)					
	P5	P4	P3	P2	P1
CABLE P/N	655-185-03	655-185-03	655-185-03	655-181-03	655-181 -03

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

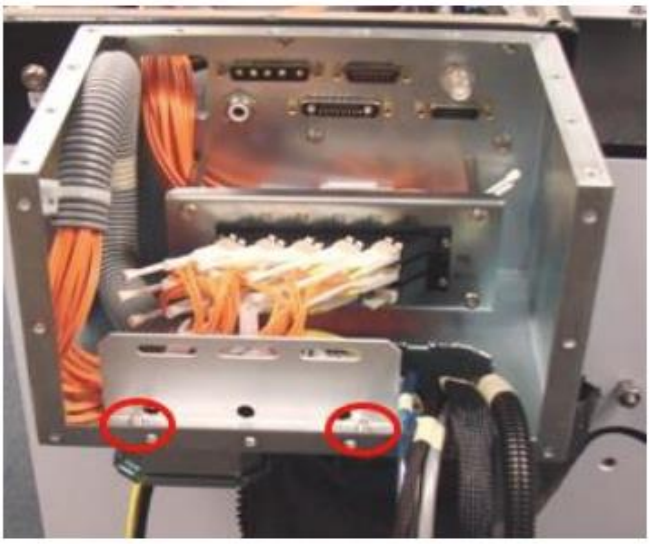
MACHINE:

PH2 Phillips screwdriver, 3.3 Nm

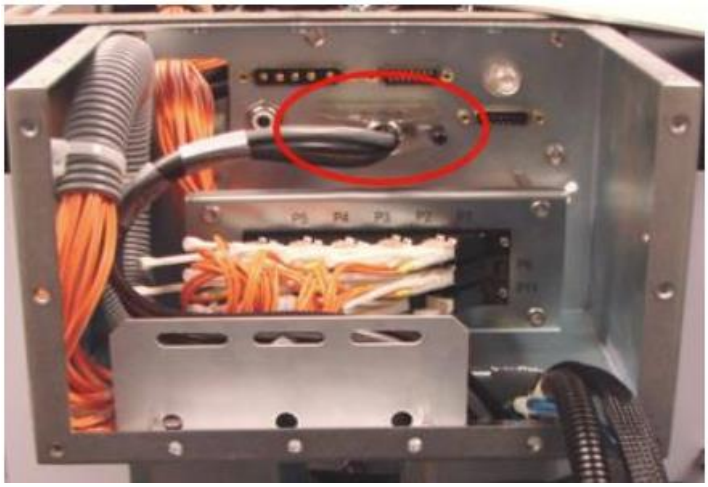
METHOD:

h. Reinstall the previously saved front bracket of the circuit breaker panel, using the two saved M5 screws. The tool is a PH2 Phillips screwdriver with a torque value of 3.3 Nm.

Pay attention to check that the signal line is not pinched!



i. Install the big "J2" connector as shown in the figure on the matching port of the disconnect panel and tighten it manually.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
49	813-929-01	HDW VELCRO STRP 5/8	2

MACHINE:

flat-blade screwdriver 0.19 N·m

METHOD:

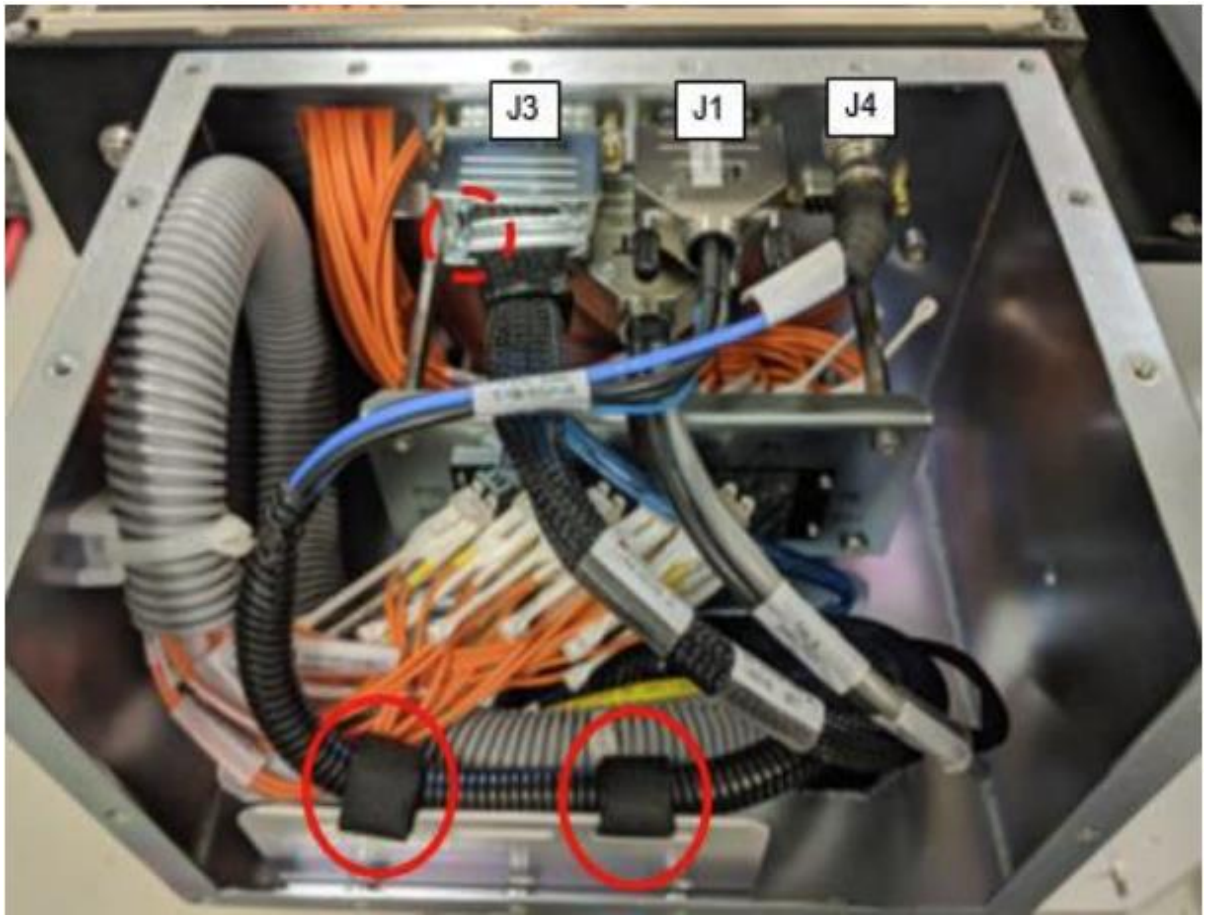
j. Route and connect the transparent blue air supply hose as shown in the figure to the corresponding air pipe port on the broken pavement plate under the connector J3.

k. Connect the wires to the matching connector panel in the following order as shown in the figure.

J4 and J1 interfaces are tightened manually, J3 interface is used

Tool flat-blade screwdriver, torque value is 0.19 N·m.

l. Use 2 cable ties (813-929-01) to tie all cables to the front bracket.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

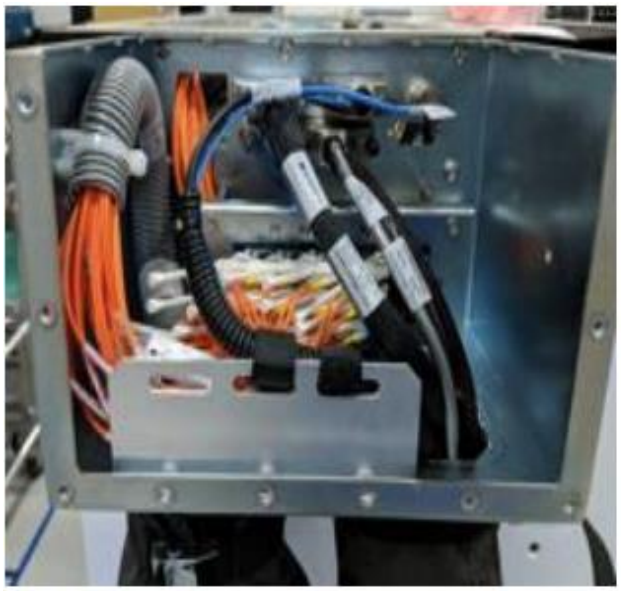
ITEM	P/N	DESCRIPTION	QTY
50	630-258-40	ELEC-MECH EMI CBL CVR SIGNAL BLACK FOR SEMI TEST	1

MACHINE:

PH2 Phillips torque screwdriver 0.7 N·m

METHOD:

25.1 T/H to the cabinet assembly-install the EMI cable sleeve on the panel.
b. Install the cover as shown in the figure and fix it with the 20 screws removed before.
The tool is a PH2 Phillips torque screwdriver with a torque of 0.7 N·m.
These screws will be removed when button up and reinstalled with standard torque.
c. As shown in the picture, pull up the zipper and tighten the buckle on the cable sleeve to fix the cable sleeve on the panel.



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
51	661-717-00	PLASTIC, UFPLUS SYSTEM MODEL, LABEL	1
52	479-133-42	CHINA ROHS LABEL	1

MACHINE:

N/A

METHOD:

26.1 As shown in the figure, bag 1 system label (661-717-00) and 1 RoHS label (479-133-42) and hang it and it will flow with the system to the next station for TE.

System label (661-717-00)

RoHS label (479-133-42)

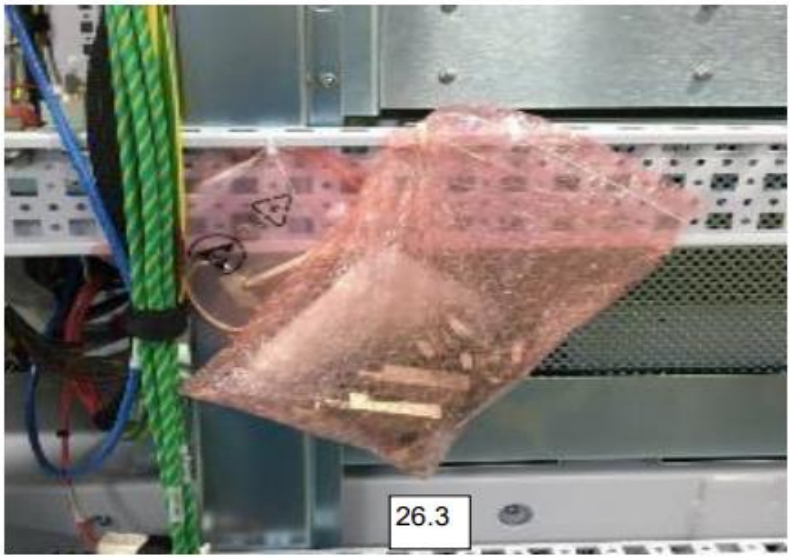
MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

26.2 Screw the 4 feet to the top as shown in the figure, confirm whether a piece of iron plate fixture can be inserted under the feet, if not, notify the PE engineer to solve it immediately!
26.3 If it can be inserted, remove the 4 studs and 4 flat feet, put the 4 361-081-50 back in the original packaging and 4 637-160-00 studs into the bubble bag and hang it
Flow up the cabinet to the next stop!



MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: **N/A**

MACHINE:

Polyester, 40mm, 10mm, white,N;Label (FLX-PKG-LBL-H00081)

METHOD:

21.4 Paste a system label with the same serial number on the cabinet and the machine head as shown in the figure.

MAN: :CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Matters needing attention

1. Take ESD and protective safety measures.
2. Be careful not to scratch the surface of the parts during the assembly process.
3. After assembly, check that the assembled parts have no defects and no missing operations.